Universality for orthoscheme reflection groups: the right-angled Coxeter envelope and the collision depth

Vico Bonfioli*

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Abstract

The $n+1$ facet reflections of an n-dimensional right-corner orthoscheme $O(a_1,\dots,a_n)$ generate a group $W(a) \leq \mathrm{Isom}(\mathbb{R}^n)$, and $W(a)$ is a quotient of the right-angled Coxeter group W_n on the orthogonality graph of the facets. This paper determines what the quotient can do to the growth, and states precisely where that determination stops.

Four statements are proved unconditionally. First, a dimension-free dichotomy: if $Q = G/N$ is a quotient of a marked group and $K(N)$ is the length of a shortest nontrivial element of N, then the sphere sizes of Q and of G agree in every degree below $\lceil K(N)/2 \rceil$ and Q is strictly smaller in that degree (Theorem 7). Second, the growth series of W_n is rational, equal to $(1+t)^{\lfloor n/2 \rfloor + 1}/K_{n+1}(t)$ for an explicit integer polynomial K_{n+1}, with growth rate $r_n = 1 + 2\cos\big(2\pi/(n+3)\big)$ (Theorem 14). Third, for $n \geq 3$ and every positive leg tuple, every element of $\ker \rho_a$ of word length 6 is $(R_i R_{i+1})^{\pm 3}$ for a pair of consecutive facets meeting at angle $\pi/3$ (Theorem 11). Fourth, a complete classification of the rank-two relations for rational leg tuples: for $a \in \mathbb{Q}^n_{>0}$, $\ker(W_n \to W(a))$ meets a rank-two standard parabolic nontrivially if and only if the leg sequence has two equal legs at an endpoint of the staircase or three consecutive equal legs (Theorem 23). The rationality hypothesis is not removable: at the real tuple $a = (\sqrt{3}, 1, 5)$ the endpoint pair closes at angle $\pi/3$ with no two legs equal (Remark 24). The Diophantine input to the fourth is that three quartics in the flanking legs have only trivial rational points, their Jacobians being the rank-zero curves `24a1`, `72a2` and $y^2 = x^3 - x$.

Together these determine the collision depth $\mathrm{cd}_n(a) = \inf\{d : u_d(a) < [t^d]\, W_n(t)\}$, which equals $K(\ker \rho_a)/2$ and is at least 3 for every positive leg tuple, from the two positional statistics (e,t) of the leg sequence, again for $a \in \mathbb{Q}^n_{>0}$: $\mathrm{cd}_n = 3$ when three consecutive legs are equal, $\mathrm{cd}_n = 4$ when only an endpoint pair is equal, and on the remaining stratum $e = t = 0$, $\mathrm{cd}_n = \infty$ exactly when ρ_a is injective and $\mathrm{cd}_n \geq 4$ otherwise (Theorem 29). The first two cases are unconditional.

Whether ρ_a is injective at all is decided here in dimension two only, where it is not. That boundary is declared, not incidental. The planar proof is an amenability argument, and Theorem 34 shows that it cannot be run in any dimension $n \geq 3$: $O(n)$ contains a free subgroup of rank two for $n \geq 3$, so $\mathrm{Isom}(\mathbb{R}^n)$ is not amenable as an abstract group, and any argument of that shape would have to prove the strictly stronger statement that W_n admits no embedding at all into $\mathrm{Isom}(\mathbb{R}^n)$. Section 8 states the three problems that remain, with what is known about each. We also record that nonvanishing of the Schur-complement determinant $\det Q_n = -\prod_i a_i^2$ (Lemma 26) does not imply faithfulness, contrary to an argument used in an earlier version of this material (Remark 27).

*Independent researcher, Italy. `vicobonfioli@gmail.com`. https://elvec1o.github.io/home/posts/.

1. Introduction

An *n-dimensional right-corner orthoscheme*, or path simplex, $O = O(a_1, \dots, a_n) \subset \mathbb{R}^n$ is the simplex with vertices

$$V_0 = 0, \qquad V_k = V_{k-1} + a_k e_k \quad (1 \le k \le n),$$

where $e_1, \dots, e_n$ is the standard basis and $a = (a_1, \dots, a_n)$ is a tuple of positive reals, the *legs*. Its $n+1$ facets carry reflections $\widehat{R}_0, \dots, \widehat{R}_n \in \mathrm{Isom}(\mathbb{R}^n)$, and we write $W(a) = \langle \widehat{R}_0, \dots, \widehat{R}_n \rangle$ for the group they generate.

Two facets $\widehat{R}_i, \widehat{R}_j$ with $|i-j| \ge 2$ are perpendicular for every leg tuple (Lemma 1), while consecutive facets meet at an angle that depends on the legs and is generically not a rational multiple of π. The largest group compatible with the leg-independent relations is therefore the right-angled Coxeter group

$$W_n = \langle R_0, \dots, R_n \mid R_i^2 = 1,\ (R_i R_j)^2 = 1\ (|i-j| \ge 2) \rangle, \tag{1}$$

on the *orthogonality graph* Γ_n (vertices $0, \dots, n$, an edge whenever $|i-j| \ge 2$), and there is a surjection $\rho_a \colon W_n \twoheadrightarrow W(a)$ for every a. Distinguishing W_n from $W(a)$ is essential: writing W_n for the geometric group would make "ρ_a is faithful" vacuous, and it is exactly the possible failure of injectivity that this paper is about.

1.1 Results

The sphere sizes $u_d(a) = \#\{w \in W(a) : \ell(w) = d\}$ are bounded above by the coefficients of the growth series $W_n(t)$ of the envelope (1), and the whole question is where and by how much the bound fails to be sharp. Four unconditional theorems answer the "where".

- **The dichotomy** (Theorem 7). For any group G with a finite symmetric generating set and any $N \trianglelefteq G$ with $N \ne \{1\}$, put $K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$. Then the sphere sizes of G/N agree with those of G in every degree $d < \lceil K(N)/2 \rceil$, and are strictly smaller in degree $\lceil K(N)/2 \rceil$. So a family of quotients of a fixed G has a universal initial segment, and the horizon of universality is half the shortest added relator.

- **The envelope** (Theorem 14). $W_n(t)$ is rational, and is $(1+t)^{\lfloor n/2 \rfloor + 1}/K_{n+1}(t)$ for the explicit integer polynomials K_m of (2). Its growth rate is $r_n = 1 + 2\cos(2\pi/(n+3))$.

- **The rank-two classification** (Theorem 23). For $a \in \mathbb{Q}^n_{>0}$, the kernel of ρ_a meets a rank-two standard parabolic $\langle R_i, R_{i+1} \rangle$ nontrivially if and only if $e(a) + t(a) \ge 1$, where $e(a)$ counts equal legs at the two endpoints of the staircase and $t(a)$ counts three consecutive equal legs. The shortest such element has length 6 when $t(a) \ge 1$ and length 8 when $t(a) = 0 < e(a)$.

- **Length six is dihedral** (Theorem 11). For $n \ge 3$ and every $a \in U$, an element of $\ker \rho_a$ of word length 6 is $(R_i R_{i+1})^{\pm 3}$ for a consecutive pair of facets meeting at angle $\pi/3$. There is no length-six relation outside the rank-two standard parabolics, in any dimension $n \ge 3$ and at any positive leg tuple.

The first two combine into the statement that gives the paper its shape. Define the *collision depth* (Definition 4)

$$\mathrm{cd}_n(a) = \inf\{d \ge 0 :\ u_d(a) < [t^d]\, W_n(t)\} \in \{1, 2, \dots\} \cup \{\infty\}.$$

By the dichotomy, $\mathrm{cd}_n(a) = \lceil K(\ker \rho_a)/2 \rceil$, and since every element of the kernel has even length this is $K(\ker \rho_a)/2$ (Corollary 9). A short geometric argument then shows $K(\ker \rho_a) \ge 6$ for every positive leg tuple (Lemma 10), so $\mathrm{cd}_n \ge 3$ unconditionally. The last two theorems pin the value: $K(a) = 6$ if and only if three consecutive legs are equal (Corollary 28), so $\mathrm{cd}_n = 3$ exactly on that stratum, $\mathrm{cd}_n = 4$ when only an endpoint pair is equal, and $\mathrm{cd}_n \geq 4$ everywhere else (Theorem 29). The remaining case $e = t = 0$ is $\mathrm{cd}_n = \infty$ precisely when ρ_a is injective.

1.2 Scope

The four theorems above are unconditional; they hold at every positive leg tuple apart from the arithmetic hypotheses of Theorem 23, and they are dimension-free except that Theorem 11 requires $n \geq 3$. That restriction is a restriction on the proof; at $n = 2$ the conclusion is left undecided here, in either direction (Remark 13). The question the four do not answer is whether ρ_a is injective at all when $e = t = 0$, and this paper answers that question in dimension two only, where the answer is no (Proposition 32). That is the declared boundary of the paper, and it is a boundary of the available method rather than of the write-up.

Theorem 34 makes that last sentence precise. The planar proof is an amenability argument: $\mathrm{Isom}(\mathbb{R}^2)$ is virtually solvable, hence amenable, and so is every subgroup of it, whereas W_2 contains a free subgroup of rank two and is not. For $n \geq 3$ that argument is not merely unproved but unavailable, since $SO(3) \leq SO(n)$ contains a free subgroup of rank two and $\mathrm{Isom}(\mathbb{R}^n)$ is therefore not amenable as an abstract group. Theorem 34 also identifies what any argument of that shape would have to establish, namely that W_n admits no embedding whatsoever into $\mathrm{Isom}(\mathbb{R}^n)$, a statement strictly stronger than non-injectivity of the particular map ρ_a.

Section 8 states the three problems that remain: faithfulness on the $e = t = 0$ stratum (Conjecture 25, with the generic and arithmetic forms of Problem 36), the embedding question just described (Problem 37), and finite presentation of the generic n-orthoscheme group (Problem 38). Each is stated with what is known about it. Nothing outside those three is left implicit: Table 3 assigns a status to every numbered statement in the paper.

1.3 What is and is not new

The dichotomy is elementary. It is recorded here with a proof because it is what turns the two halves of this paper into statements about each other, and because its deviation half needs a case distinction on the parity of $K(N)$ that is easy to omit; but it is not a difficult argument, and the phenomenon it describes, that a family of quotients of a common group shares an initial segment of its growth, is a general one of which the orthoscheme family is an instance. The present paper claims the instance, not the principle.

Both ingredients of r_n are classical and are used as such. Steinberg's formula for the Poincaré series of a Coxeter group is standard [21, 13, 3], growth series of right-angled Coxeter groups are treated systematically by Kellerhals and Perren [15], and the independence polynomials of paths, their Binet formula, and the location of their roots are classical facts about the Fibonacci–Pell family. What is asserted is the identification: that the growth rate of the n-dimensional right-corner orthoscheme envelope is the largest of those roots at $m = n+3$. We have not located that identification in the literature and make no claim of novelty for the ingredients. A reader who knows the Coxeter growth-series literature should read Theorem 14 as an assembly of known pieces.

The Diophantine part of Theorem 23 rests on three rank-zero Mordell–Weil computations that are in the standard tables [9, 16, 17] and have been recomputed here (Remark 21); what is done here is the reduction of the dihedral condition to those three curves.

Two points of contact should be recorded. The all-equal tuple $a_1 = \cdots = a_n$ is not a generic point of this family: there the orthoscheme group is the affine Weyl group $\widetilde{C}_n$, whose growth is classical and whose Poincaré series is given by Bott's formula, and which reproduces the $\mathrm{cd}_n = 3$ rows of Table 2. In the planar case $n = 2$, Isola and Piergallini [14] prove that the generic triangle reflection group is not finitely presented, which already implies that ρ_a is not injective there; that is a second reason, besides the amenability argument of Proposition 32, why the plane is not a guide to $n \geq 3$.

1.4 Companion papers

The planar case $n = 2$ is treated separately [2]: there the metric has a lamplighter-type closed form, the deviation from the envelope has been computed past its onset, and the resulting growth series (A396406, growth rate $\beta_2 = 1.4916\ldots$) is transcendental over $\mathbb{Q}(x)$. Nothing below is quoted from that work except the two numerical facts recorded in Corollary 33, and the transcendence questions, which are special to $n = 2$, are not attempted here. Non-right-corner orthoschemes are outside the scope of this paper.

2. Orthoschemes and their Coxeter envelope

Throughout, $n \geq 2$, and

$$U = \{a = (a_1, \ldots, a_n) \in \mathbb{R}^n : \ a_i > 0 \text{ for all } i\}$$

is the leg space. Everything below is invariant under simultaneous scaling of the legs, so one may normalise if convenient; we do not.

2.1 Facet normals

Lemma 1 (Facet normals)**.** *With $V_0 = 0$ and $V_k = \sum_{i \leq k} a_i e_i$, the facet of $O(a)$ opposite V_j has normal*

$$m_0 = e_1, \qquad m_j = a_{j+1}e_j - a_j e_{j+1} \ \ (1 \leq j \leq n-1), \qquad m_n = e_n.$$

Consequently $\operatorname{supp}(m_0) = \{1\}$, $\operatorname{supp}(m_j) = \{j, j+1\}$ *and* $\operatorname{supp}(m_n) = \{n\}$, *so $m_i \cdot m_j = 0$ whenever $|i - j| \geq 2$, while $m_i \cdot m_{i+1} \neq 0$ for every i and every $a \in U$.*

Proof. The facet opposite V_j is the affine hull of $\{V_i : i \neq j\}$. For $j = 0$ the difference vectors $V_{k+1} - V_k = a_{k+1}e_{k+1}$, $1 \leq k \leq n-1$, span $\langle e_2, \ldots, e_n\rangle$, whence $m_0 = e_1$. For $j = n$ the vertices $V_0, \ldots, V_{n-1}$ span $\langle e_1, \ldots, e_{n-1}\rangle$, whence $m_n = e_n$. For $1 \leq j \leq n-1$ the difference vectors are $a_k e_k$ for $k \leq j-1$, then $V_{j+1} - V_{j-1} = a_j e_j + a_{j+1}e_{j+1}$, then $a_k e_k$ for $k \geq j+2$. Orthogonality to the first and third families confines m_j to coordinates $\{j, j+1\}$, and orthogonality to the middle vector forces $m_j \propto a_{j+1}e_j - a_j e_{j+1}$. Two supports of the listed form are disjoint exactly when $|i - j| \geq 2$. For adjacent indices, $m_0 \cdot m_1 = a_2$, $m_j \cdot m_{j+1} = -a_j a_{j+2}$ for $1 \leq j \leq n-2$, and $m_{n-1} \cdot m_n = -a_{n-1}$, all nonzero because every $a_i > 0$. $\square$

Lemma 1 holds for all legs, with no genericity hypothesis and no case check. Hence $\widehat{R}_i^2 = 1$ always and $\widehat{R}_i\widehat{R}_j = \widehat{R}_j\widehat{R}_i$ whenever $|i - j| \geq 2$, so the assignment $R_i \mapsto \widehat{R}_i$ defines a surjection

$$\rho_a \colon W_n \twoheadrightarrow W(a) \leq \operatorname{Isom}(\mathbb{R}^n)$$

from the right-angled Coxeter group (1) on the orthogonality graph Γ_n [10]. The abstract group W_n does not depend on a.

We write H_j for the facet hyperplane carrying $\widehat{R}_j$ and $\theta_{i,j} \in (0, \frac{\pi}{2}]$ for the angle between H_i and H_j, so that

$$\cos\theta_{i,j} = \frac{|m_i \cdot m_j|}{|m_i|\,|m_j|}.$$

For $i \neq j$ the isometry $\widehat{R}_i\widehat{R}_j$ fixes $H_i \cap H_j$ and acts on the orthogonal 2-plane as a rotation through $2\theta_{i,j}$ *or* through $2\pi - 2\theta_{i,j}$. The absolute value in the display, equivalently the normalisation

$\theta_{i,j} \le \pi/2$, records the unordered pair of supplementary dihedral angles and discards the sense of the rotation, so the alternative cannot be resolved without further conventions. It is never resolved below. Only three consequences of the description are used, and each is invariant under $\varphi \mapsto 2\pi - \varphi$: the rotation angle is nonzero modulo 2π, so $\mathrm{Fix}(\widehat{R}_i\widehat{R}_j) = H_i \cap H_j$ exactly; the order of $\widehat{R}_i\widehat{R}_j$ is the order of $2\theta_{i,j}$ in $\mathbb{R}/2\pi\mathbb{Z}$; and $\widehat{R}_i\widehat{R}_j$ is the rotation by π if and only if $\theta_{i,j} = \pi/2$.

The intersections of the facet hyperplanes are recorded separately, as several later proofs use them.

Lemma 2 (Facet intersections)**.** *Let $n \ge 2$ and $a \in U$. The vertices $V_0, \dots, V_n$ are affinely independent, $H_j = \mathrm{aff}\{V_m : m \ne j\}$, and for $i \ne j$ the hyperplanes H_i and H_j are distinct, non-parallel and satisfy*

$$H_i \cap H_j = \mathrm{aff}\{V_m : m \ne i, j\},$$

a flat of dimension $n-2$. Distinct index pairs give distinct intersections.

Proof. The difference vectors $V_k - V_0 = \sum_{i \le k} a_i e_i$, $1 \le k \le n$, form a triangular matrix with diagonal $a_1, \dots, a_n$, all nonzero, so they are linearly independent and the $n+1$ vertices are affinely independent. The facet opposite V_j is by definition $\mathrm{aff}\{V_m : m \ne j\}$, a flat of dimension $n-1$, that is H_j; and $V_j \notin H_j$, again by affine independence. Hence $H_i \ne H_j$ for $i \ne j$, since $V_j \in H_i$ while $V_j \notin H_j$. The flat $F = \mathrm{aff}\{V_m : m \ne i, j\}$ has dimension $n-2$ and lies in $H_i \cap H_j$; two distinct hyperplanes meet in a flat of dimension $n-2$ or not at all, so $H_i \cap H_j = F$ and the two are not parallel. If $\mathrm{aff}\{V_m : m \ne i, j\} = \mathrm{aff}\{V_m : m \ne k, l\}$ with $\{i, j\} \ne \{k, l\}$, the common flat would contain at least n of the vertices and so have dimension at least $n-1$, a contradiction. ☐

The $n+1$ normals live in $\mathbb{R}^n$, so they carry one linear relation. It is unique and involves all of them, which is the source of every independence statement used below.

Lemma 3 (The unique relation among the normals)**.** *Let $a \in U$. The vectors $m_0, \dots, m_n$ span $\mathbb{R}^n$, and the space of linear relations $\sum_j c_j m_j = 0$ is one-dimensional, spanned by the relation determined by*

$$c_0 = -a_2 c_1, \qquad c_k = \frac{a_{k-1}}{a_{k+1}}\, c_{k-1} \quad (2 \le k \le n-1), \qquad c_n = a_{n-1} c_{n-1},$$

in which every coefficient is nonzero. Consequently any at most n of the vectors $m_0, \dots, m_n$ are linearly independent.

Proof. The matrix with rows $m_1, \dots, m_n$ is upper triangular in the basis $e_1, \dots, e_n$: row j has entries a_{j+1} in column j and $-a_j$ in column $j+1$ for $1 \le j \le n-1$, and row n is e_n. Its determinant is $a_2 a_3 \cdots a_n \ne 0$, so $m_1, \dots, m_n$ is a basis and the relation space of the $n+1$ vectors is one-dimensional. Only m_0 and m_1 involve e_1, so the e_1-coefficient of $\sum_j c_j m_j$ is $c_0 + a_2 c_1$; for $2 \le k \le n-1$ only m_{k-1} and m_k involve e_k, giving $a_{k+1} c_k - a_{k-1} c_{k-1}$; and the e_n-coefficient is $c_n - a_{n-1} c_{n-1}$. Setting these to zero gives the displayed recursion, and all a_i are positive, so $c_1 \ne 0$ forces every $c_j \ne 0$. A dependent subset of size at most n would give a nonzero relation with at least one vanishing coefficient, contradicting uniqueness up to scale. ☐

2.2 Counting conventions

For a group H with a marked finite generating set we write ℓ_H for the word length and $s_d(H) = \#\{h \in H : \ell_H(h) = d\}$ for the sphere sizes. We abbreviate

$$u_d(a) = s_d(W(a)), \qquad W_n(t) = \sum_{d \ge 0} s_d(W_n)\, t^d,$$

the generators being $\widehat{R}_0, \dots, \widehat{R}_n$ and $R_0, \dots, R_n$ respectively. Since $O(a)$ is a fundamental domain for the action of $W(a)$, $u_d(a)$ is also the number of distinct chambers $g \cdot O(a)$ at word distance d, which is what the breadth-first search of §10 enumerates. Being the sphere sizes of a quotient of W_n, they satisfy $u_d(a) \le [t^d]\, W_n(t)$ for every d.

Finally, set

$$K(a) = \min\{\ell_{W_n}(w) : w \in \ker \rho_a \setminus \{1\}\} \in \{1, 2, \dots\} \cup \{\infty\},$$

with $K(a) = \infty$ when ρ_a is injective.

Definition 4 (Collision depth)**.** For $a \in U$ the *collision depth* is

$$\mathrm{cd}_n(a) = \inf\{d \ge 0 : u_d(a) < [t^d]\, W_n(t)\} \in \{1, 2, \dots\} \cup \{\infty\},$$

with $\mathrm{cd}_n(a) = \infty$ when the set is empty. The definition is made at every positive leg tuple; rationality of the legs is a hypothesis of some statements about cd_n in §6, never of the definition.

2.3 Generic rigidity

Before computing anything we record that "the generic orthoscheme group" is a well-defined object: it does not depend on which generic shape one picks. Write $N_{\mathrm{gen}} = \bigcap_{a\in U} \ker \rho_a \trianglelefteq W_n$.

Theorem 5 (Generic rigidity)**.** *For every $n \ge 2$ there is a co-null subset $U_{\mathrm{gen}} \subseteq U$, the complement of a countable union of proper real-algebraic subvarieties, such that $\ker \rho_a = N_{\mathrm{gen}}$, hence $W(a) \cong W_n/N_{\mathrm{gen}}$, for every $a \in U_{\mathrm{gen}}$. Consequently $u_d(a)$ is constant on U_{gen}, equal to $s_d(W_n/N_{\mathrm{gen}})$, and this generic value is the maximum of u_d over all of U. Moreover U_{gen} contains every a whose coordinates are algebraically independent over $\mathbb{Q}$; in particular it is dense.*

Proof. Each facet normal $m_j(a)$ is a fixed polynomial vector in a by Lemma 1, so the reflection $\widehat{R}_j = I - 2\, m_j m_j^{\top}/(m_j \cdot m_j)$, together with its translation part, has entries in $\mathbb{Q}(a)$ with denominator the nowhere-vanishing polynomial $m_j \cdot m_j$. Hence for each $w \in W_n$, choosing any word representative, every entry of $\rho_a(w)$ lies in $\mathbb{Q}(a)$, independently of the representative because ρ_a is well defined. The locus $V_w = \{a \in U : \rho_a(w) = I\}$ is the simultaneous vanishing of finitely many rational functions cleared of their nonvanishing denominators, hence Zariski-closed in the irreducible variety U. If $w \notin N_{\mathrm{gen}}$ then $\rho_a(w) \ne I$ for some a, so V_w is a proper closed subvariety, of measure zero. As W_n is countable, $B := \bigcup_{w \in W_n \setminus N_{\mathrm{gen}}} V_w$ is a countable union of proper subvarieties, and $U_{\mathrm{gen}} := U \setminus B$ is co-null and dense.

For $a \in U_{\mathrm{gen}}$ one has $N_{\mathrm{gen}} \subseteq \ker \rho_a$ by definition, and $\ker \rho_a \subseteq N_{\mathrm{gen}}$ because any $w \in \ker \rho_a \setminus N_{\mathrm{gen}}$ would place a in $V_w \subseteq B$. Isomorphic groups with marked generators have identical sphere sizes, giving the constant value; for arbitrary a, $\ker \rho_a \supseteq N_{\mathrm{gen}}$ makes $W(a)$ a quotient of W_n/N_{gen}, so $u_d(a) \le s_d(W_n/N_{\mathrm{gen}})$. Finally each V_w is defined over $\mathbb{Q}$; if the coordinates of a are algebraically independent over $\mathbb{Q}$ then a lies on no proper $\mathbb{Q}$-subvariety, so $a \notin B$. Such a exist since $\dim U = n \ge 2$. □

Corollary 6 (One faithful shape suffices)**.** *If ρ_a is injective for a single $a \in U$, then $N_{\mathrm{gen}} = \{1\}$ and ρ_b is injective for every $b \in U_{\mathrm{gen}}$. Conversely, if ρ_b fails to be injective for some $b \in U_{\mathrm{gen}}$, then ρ_a fails to be injective for every $a \in U$.*

Proof. $N_{\mathrm{gen}} \subseteq \ker \rho_a = \{1\}$, and $\ker \rho_b = N_{\mathrm{gen}}$ for $b \in U_{\mathrm{gen}}$ by Theorem 5. The converse is the contrapositive, using $N_{\mathrm{gen}} \subseteq \ker \rho_a$ for every a. □

Corollary 6 is what allows the generic question and the arithmetic question of §8 to be treated as two forms of one problem in one direction. It gives nothing in the other direction: a specific rational tuple need not lie in U_{gen}, and U_{gen} contains no tuple with algebraically dependent coordinates, hence no rational tuple at all.

3. The universality–deviation dichotomy

Theorem 7 (Universality–deviation dichotomy)**.** *Let $G = \langle S \rangle$ be a group with finite symmetric generating set S, let $N \trianglelefteq G$ with $N \neq \{1\}$, write $Q = G/N$ and $K = K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$, and put $h = \lceil K/2 \rceil$. Then $s_d(Q) = s_d(G)$ for every $d < h$, and $s_h(Q) < s_h(G)$.*

Proof. Write $\pi : G \to Q$.

Agreement below h. Let $d < h$. If $d \geq 1$ then $2d \leq 2h - 2 \leq K - 1 < K$, since $2h \leq K + 1$. Suppose $g, g' \in B_d(G)$ satisfy $\pi g = \pi g'$. Then $g(g')^{-1} \in N$ and $\ell_G(g(g')^{-1}) \leq 2d < K$, so $g = g'$; thus π is injective on $B_d(G)$. It is also length-preserving there: if $\ell_G(g) = d$ and $\ell_Q(\pi g) = e < d$, pick g' with $\ell_G(g') = e$ and $\pi g' = \pi g$; then $\ell_G(g(g')^{-1}) \leq d + e < 2d < K$ forces $g = g'$, contradicting $e < d$. Finally π maps the sphere of radius d in G onto the sphere of radius d in Q, because $\ell_Q(q) = \min\{\ell_G(g) : \pi g = q\}$ realises every q with $\ell_Q(q) = d$ by some g of length exactly d. Hence $s_d(Q) = s_d(G)$.

Strict drop at h. Choose $g_0 \in N \setminus \{1\}$ with $\ell_G(g_0) = K$ and split a geodesic word for it as $g_0 = w_1 w_2$ with $\ell_G(w_1) = h$ and $\ell_G(w_2) = K - h = \lfloor K/2 \rfloor$; both factors are geodesic as written. From $w_1 w_2 \in N$ we get $\pi(w_1) = \pi(w_2^{-1})$, and $w_1 \neq w_2^{-1}$ since otherwise $g_0 = 1$. Surjectivity of π from the radius-h sphere of G onto that of Q, established in the previous paragraph for $d < h$, holds verbatim for $d = h$.

If K is even then $h = K/2 = \lfloor K/2 \rfloor$, so w_1 and w_2^{-1} are two distinct elements of the sphere of radius h in G with the same image, and $s_h(Q) \leq s_h(G) - 1$.

If K is odd then $h = (K+1)/2$ and $\ell_G(w_2^{-1}) = h - 1$, so $\ell_Q(\pi w_1) \leq h - 1 < h$ and the image of w_1 does not lie on the sphere of radius h in Q. That sphere is therefore covered by π applied to the radius-h sphere of G with w_1 removed, and again $s_h(Q) \leq s_h(G) - 1$. ☐

The parity split at the end is not cosmetic. When K is odd the two halves of a shortest relator lie on spheres of different radii, so no two points of the upstairs sphere need collide, and the count drops for the other reason. Both cases are machine-checked (§10).

Applied to ρ_a, Theorem 7 gives an exact alternative with no genericity and no dimension hypothesis.

Corollary 8 (The envelope alternative)**.** *Exactly one of the following holds.*

(i) ρ_a is injective. Then $W(a) \cong W_n$ and $u_d(a) = [t^d]\, W_n(t)$ for every d; in particular the growth series of $W(a)$ is the rational function of Theorem 14, with growth rate r_n.

(ii) ρ_a is not injective. Then $K(a) < \infty$, $u_d(a) = [t^d]\, W_n(t)$ for every $d < \lceil K(a)/2 \rceil$, and $u_d(a) < [t^d]\, W_n(t)$ at $d = \lceil K(a)/2 \rceil$.

Proof. If $\ker \rho_a = \{1\}$ then ρ_a is an isomorphism of marked groups and (i) is immediate. Otherwise $K(a)$ is finite and Theorem 7 with $G = W_n$, $N = \ker \rho_a$ gives (ii). ☐

Corollary 9 (The collision depth is half the shortest relator)**.** *For every $a \in U$, every element of $\ker \rho_a$ has even word length in W_n. Consequently $\mathrm{cd}_n(a) = \lceil K(a)/2 \rceil = K(a)/2$, where $K(a)/2$ is read as ∞ when $K(a) = \infty$.*

Proof. Parity. Let $\varepsilon : W_n \to \{\pm 1\}$ be the sign character, $\varepsilon(R_i) = -1$, which is well defined because all defining relators of (1) have even length; then $\varepsilon(w) = (-1)^{\ell(w)}$. On the other side, $\det \circ \rho_a$ is a homomorphism $W_n \to \{\pm 1\}$ sending each R_i to -1, so $\det \rho_a(w) = \varepsilon(w)$. If $w \in \ker \rho_a$ then $\det \rho_a(w) = 1$, hence $\ell(w)$ is even. In particular $K(a)$ is even whenever it is finite. The statement that $K(a)$ itself is even is available only in that case: $K(a) = \infty$ is not an even number.

The equality. If $\ker\rho_a = \{1\}$ then $K(a) = \infty$ and, by Corollary 8(i), the set in Definition 4 is empty, so both sides are ∞. Otherwise $K(a)$ is finite and Corollary 8(ii) gives agreement below $\lceil K(a)/2\rceil$ and strict inequality at $\lceil K(a)/2\rceil$, so $\mathrm{cd}_n(a) = \lceil K(a)/2\rceil$, which equals $K(a)/2$ by parity. $\square$

Lemma 10 (No relator of length four)**.** *For every $n \geq 2$ and every $a \in U$, $K(a) \geq 6$. Consequently $\mathrm{cd}_n(a) \geq 3$.*

Proof. By Corollary 9 every element of $\ker\rho_a$ has even length, so it suffices to exclude $K(a) \in \{2, 4\}$.

By Lemma 2 the facet hyperplanes H_i are pairwise distinct and pairwise non-parallel, $H_i \cap H_j = \mathrm{aff}\{V_m : m \neq i, j\}$ has dimension $n-2$, and distinct index pairs give distinct intersections.

$K(a) = 2$ would give a geodesic word R_iR_j with $i \neq j$ and $\widehat{R}_i = \widehat{R}_j$, impossible since $H_i \neq H_j$.

Suppose $K(a) = 4$, realised by a geodesic word $R_iR_jR_kR_l$ with $i \neq j$, $j \neq k$, $k \neq l$ and $\widehat{R}_i\widehat{R}_j\widehat{R}_k\widehat{R}_l = 1$, that is $\widehat{R}_i\widehat{R}_j = \widehat{R}_l\widehat{R}_k$. For $H_i \neq H_j$ non-parallel, the isometry $\widehat{R}_i\widehat{R}_j$ is a rotation about $H_i \cap H_j$ through a nonzero angle, whichever of the two supplementary readings of §2 is taken; its fixed-point set is therefore exactly $H_i \cap H_j$. The same applies to $\widehat{R}_l\widehat{R}_k$, so $H_i \cap H_j = H_k \cap H_l$ and hence $\{i, j\} = \{k, l\}$ by the previous paragraph. Since $j \neq k$ this forces $i = k$ and $j = l$, so the relation reads $\widehat{R}_i\widehat{R}_j = \widehat{R}_j\widehat{R}_i$, that is $H_i \perp H_j$. For $|i-j| \geq 2$ that relation already holds in W_n and the word $R_iR_jR_iR_j$ is not geodesic; for $|i-j| = 1$ perpendicularity contradicts $m_i \cdot m_{i+1} \neq 0$ in Lemma 1. Both branches are impossible, so $K(a) \neq 4$. $\square$

3.1 Length six is dihedral

The next value, $K(a) = 6$, is the first that occurs. The point of this subsection is that when it occurs it occurs for the obvious reason: a length-six kernel element is always a dihedral relator inside a rank-two standard parabolic, in every dimension $n \geq 3$ and at every positive leg tuple. Nothing arithmetic enters; the proof is the rank calculus of products of reflections together with the affine geometry of the one facet hyperplane that misses the origin.

We use the following standard fact, in the form given by Carter [5, Lemma 2]. Let V be a Euclidean space and $g = s_{\alpha_1}\cdots s_{\alpha_k}$ a product of reflections in the hyperplanes $\alpha_i^\perp$. Then $\mathrm{im}(g-1) \subseteq \langle\alpha_1, \ldots, \alpha_k\rangle$ always, and if $\alpha_1, \ldots, \alpha_k$ are linearly independent then $\mathrm{rank}(g-1) = k$, so that $\mathrm{im}(g-1)$ is the whole of $\langle\alpha_1, \ldots, \alpha_k\rangle$ and $\mathrm{Fix}(g) = \bigcap_i \alpha_i^\perp$. We also use twice that $\mathrm{im}(g-1) = \mathrm{Fix}(g)^\perp = \mathrm{Fix}(g^{-1})^\perp = \mathrm{im}(g^{-1}-1)$ for g orthogonal.

Theorem 11 (Length-six kernel elements are dihedral)**.** *Let $n \geq 3$ and $a \in U$. Every $w \in \ker\rho_a$ with $\ell_{W_n}(w) = 6$ lies in a rank-two standard parabolic. Explicitly, $w = (R_iR_{i+1})^{\pm 3}$ for an index i with $\theta_{i,i+1} = \pi/3$.*

Proof. Write $L\colon \mathrm{Isom}(\mathbb{R}^n) \to O(n)$ for the linear part and $s_j = L(\widehat{R}_j)$ for the reflection in $m_j^\perp$. The vertex $V_0 = 0$ lies on every facet except the one opposite it, so $H_1, \ldots, H_n$ contain the origin while $H_0 = \{x_1 = a_1\}$ does not; accordingly $\widehat{R}_j(x) = s_jx$ for $1 \leq j \leq n$ and $\widehat{R}_0(x) = s_0x + 2a_1e_1$.

Fix a reduced word $w = R_{i_1}\cdots R_{i_6}$ representing an element of $\ker\rho_a$.

(0) Rotations, and the reduction to large support. For a factorisation $w = xy$ the element $yx = x^{-1}wx$ lies in $\ker\rho_a$ and is nontrivial in W_n, since $yx = 1$ would give $y = x^{-1}$ and $w = 1$. Hence $\ell(yx) \geq K(a) \geq 6$ by Lemma 10, while $\ell(yx) \leq 6$; so every cyclic rotation of the word is again a reduced word of length 6 representing an element of $\ker\rho_a$, with the same support. We may therefore rotate freely. If $|\mathrm{supp}(w)| \leq 2$ then w lies in a standard parabolic of rank at most two, which is the assertion; so assume $|\mathrm{supp}(w)| \geq 3$ and derive a contradiction.

(1) A repeated letter at distance two forces the dihedral case. Suppose $i_k = i_{k+2}$ for some k,

indices read cyclically. Rotate so that $k = 1$ and put $a' = i_1 = i_3$, $b = i_2$. Reducedness forces $|a' - b| = 1$, since $R_{a'} R_b R_{a'} = R_b$ when the two generators commute. Let $u = R_{a'} R_b R_{a'}$ and $v = R_{i_4} R_{i_5} R_{i_6}$. Then $L\rho_a(u)$ is the reflection in $s_{a'}(m_b)^\perp$, so $\operatorname{rank}(L\rho_a(u) - 1) = 1$; since $L\rho_a(v) = L\rho_a(u)^{-1}$ is orthogonal, $\operatorname{rank}(L\rho_a(v) - 1) = 1$ too. Were i_4, i_5, i_6 pairwise distinct, their normals would be independent by Lemma 3 (here $3 \le n$ is used) and Carter's lemma would give rank 3. Hence $i_4 = i_6$; write $c = i_4$, $d = i_5$, so $|c - d| = 1$ as before.

Now $\rho_a(u)$ and $\rho_a(v)$ are the reflections in the affine hyperplanes $\widehat{R}_{a'}(H_b)$ and $\widehat{R}_c(H_d)$, and $\rho_a(u)\rho_a(v) = 1$ with both factors involutions forces the two hyperplanes to coincide; call the common hyperplane P. As $\widehat{R}_{a'}$ fixes $H_{a'}$ pointwise, $P \supseteq H_{a'} \cap H_b$, and likewise $P \supseteq H_c \cap H_d$. Using $H_i \cap H_j = \operatorname{aff}\{V_m : m \ne i, j\}$ from Lemma 2, P contains every vertex V_m with $m \notin \{a', b\} \cap \{c, d\}$. If that intersection is empty, P contains all $n + 1$ vertices, which affinely span $\mathbb{R}^n$ and lie in no hyperplane. If it is a single index j, then P contains the n affinely independent vertices V_m, $m \ne j$, so $P = H_j$; the case $j = a'$ gives $H_b = \widehat{R}_{a'}(H_{a'}) = H_{a'}$, excluded, and the case $j = b$ gives $\widehat{R}_{a'}(H_b) = H_b$, which makes m_b an eigenvector of $s_{a'}$ and hence, the normals being pairwise non-proportional by Lemma 3, forces $m_{a'} \perp m_b$, contradicting $|a' - b| = 1$ and Lemma 1. Therefore $\{a', b\} = \{c, d\}$. The ordered pairs differ, since $(c, d) = (a', b)$ would make $w = (R_{a'} R_b R_{a'})^2 = 1$, so $(c, d) = (b, a')$ and $w = (R_{a'} R_b)^3$, of support $\{a', b\}$, contradicting $|\operatorname{supp}(w)| \ge 3$.

So every three cyclically consecutive letters of w are pairwise distinct.

(2) Dimension at least four. Put $u = R_{i_1} R_{i_2} R_{i_3}$ and $v = R_{i_4} R_{i_5} R_{i_6}$. By (1) each triple has distinct entries, so by Lemma 3 and Carter's lemma $\operatorname{im}(L\rho_a(u) - 1) = \langle m_{i_1}, m_{i_2}, m_{i_3} \rangle$ and $\operatorname{im}(L\rho_a(v) - 1) = \langle m_{i_4}, m_{i_5}, m_{i_6} \rangle$; and these two images agree because $L\rho_a(v) = L\rho_a(u)^{-1}$. For $n \ge 4$ two different three-element subsets of $\{m_0, \dots, m_n\}$ have different spans, since an index in one and not the other would produce a linear dependence among four normals, excluded by Lemma 3. Hence $\{i_1, i_2, i_3\} = \{i_4, i_5, i_6\}$. Combined with (1): $i_4 \notin \{i_2, i_3\}$ gives $i_4 = i_1$, then $i_5 \notin \{i_3, i_4\}$ gives $i_5 = i_2$, and $i_6 = i_3$. Thus $w = (R_p R_q R_r)^2$ with p, q, r distinct.

Let $E = \langle m_p, m_q, m_r \rangle$, of dimension 3, and $g = s_p s_q s_r$. Each of the three reflections is the identity on $E^\perp$ and preserves E, and $g^2 = 1$ because $\rho_a(w) = 1$. Carter's lemma gives $\operatorname{rank}(g - 1) = 3$, so $g - 1$ vanishes on $E^\perp$ and is invertible on E; an involution of E with no nonzero fixed vector is $-\mathrm{id}_E$, so $s_p s_q s_r = -\mathrm{id}$ on E, that is $s_q s_r = -s_p$ on E. On E the right-hand side is the rotation by π about the line $\mathbb{R}m_p$, and the left-hand side is the rotation about $\langle m_q, m_r \rangle^\perp \cap E$ through twice the angle between m_q and m_r, in one of the two senses. Equality of the axes gives $m_p \perp m_q$ and $m_p \perp m_r$; equality of the angles gives $m_q \perp m_r$, the rotation by π being its own inverse, so that the sense does not enter. By Lemma 1 the indices p, q, r are then pairwise at distance at least 2, so R_p, R_q, R_r commute pairwise in W_n and $(R_p R_q R_r)^2 = 1$ there, contradicting $\ell(w) = 6$.

(3) Dimension three. Here every three of the four normals span $\mathbb{R}^3$, so the span argument of (2) is unavailable and the affine part is used instead. Let k be the number of occurrences of R_0 in w. By (1) two occurrences of one letter are at cyclic distance at least 3, and a cycle of length 6 admits at most two positions pairwise at cyclic distance at least 3; hence $k \le 2$. The translation part of $\rho_a(w)$ is $2a_1 \sum_{j : i_j = 0} P_{j-1} e_1$, where $P_m = s_{i_1} \cdots s_{i_m} \in O(3)$ and $P_0 = 1$, and it must vanish. Each summand is a unit vector and $a_1 \ne 0$, so $k \ne 1$.

If $k = 0$ then $\operatorname{supp}(w) = \{1, 2, 3\}$, and by (1) six letters drawn from a three-element set with every three cyclically consecutive ones distinct repeat with period three, so $w = (R_p R_q R_r)^2$ with $\{p, q, r\} = \{1, 2, 3\}$. The second paragraph of (2) applies verbatim with $E = \mathbb{R}^3$ and yields $m_1 \perp m_2$, contradicting Lemma 1.

If $k = 2$, rotate so that $i_1 = i_4 = 0$. Vanishing of the translation part reads $e_1 + s_0 s_{i_2} s_{i_3} e_1 = 0$; applying s_0, which negates e_1, gives $s_{i_2} s_{i_3} e_1 = e_1$. The distinct indices i_2, i_3 lie in $\{1, 2, 3\}$, so their normals are independent and Carter's lemma gives $\operatorname{Fix}(s_{i_2} s_{i_3}) = m_{i_2}^\perp \cap m_{i_3}^\perp$. Thus $e_1 = m_0$

is orthogonal to m_{i_2} and m_{i_3}, so $\{i_2, i_3\} = \{2, 3\}$ by Lemma 1. But then R_0 commutes with both in W_3, and

$$w = R_0 R_{i_2} R_{i_3} R_0 R_{i_5} R_{i_6} = R_{i_2} R_{i_3} R_0 R_0 R_{i_5} R_{i_6} = R_{i_2} R_{i_3} R_{i_5} R_{i_6}$$

has length at most 4, contradicting $\ell(w) = 6$.

Every branch is impossible, so $|\operatorname{supp}(w)| = 2$ and w lies in a standard parabolic $\langle R_i, R_j \rangle$, $i \neq j$. If $|i - j| \geq 2$ that parabolic is $(\mathbb{Z}/2)^2$ and has no element of length 6, so $|i - j| = 1$ and the parabolic is infinite dihedral; its two elements of length 6 are $(R_i R_{i+1})^3$ and its inverse. Then $\rho_a(R_i R_{i+1})$ has order dividing 3, and is not the identity because $H_i \neq H_j$ by Lemma 2, so it has order exactly 3. By the order statement of §2 the class of $2\theta_{i,i+1}$ then has order 3 in $\mathbb{R}/2\pi\mathbb{Z}$, and $2\theta_{i,i+1} \in (0, \pi]$ leaves $2\theta_{i,i+1} = 2\pi/3$, that is $\theta_{i,i+1} = \pi/3$. □

Remark 12 (Where each hypothesis is used). Positivity of the legs enters at four places, not two.

(a) Nondegeneracy of $O(a)$, that is affine independence of $V_0, \ldots, V_n$, through Lemma 2. This is used inside Lemma 10, hence in step (0), and again directly in step (1) and in the final paragraph.

(b) The nonvanishing $m_i \cdot m_{i+1} \neq 0$ of Lemma 1. This closes a branch in each of steps (1), (2) and (3), and one inside Lemma 10.

(c) The independence statement of Lemma 3, used in steps (1), (2) and (3), and in step (1) also in the form that the normals are pairwise non-proportional.

(d) The inequality $a_1 \neq 0$ in step (3), which is what makes a single occurrence of R_0 leave a nonzero translation.

Each of the four uses only that the legs are nonzero, never their sign or their order, so Theorem 11 holds verbatim for every $a \in (\mathbb{R} \setminus \{0\})^n$. That extension is not new information: the diagonal orthogonal map $T = \operatorname{diag}(\operatorname{sgn} a_1, \ldots, \operatorname{sgn} a_n)$ carries $O(|a_1|, \ldots, |a_n|)$ to $O(a)$, facet by facet and in the same indexing, so $\rho_a = \iota_T \circ \rho_{|a|}$ with ι_T conjugation by T, and the two kernels coincide. No rationality is assumed anywhere in Theorem 11.

The hypothesis $n \geq 3$ is used in step (1), where Lemma 3 is applied to three normals at once, and again in the split between steps (2) and (3), which is the split between $n \geq 4$ and $n = 3$.

Remark 13 (The case $n = 2$). At $n = 2$ the proof above breaks down and the statement is left undecided; the two should not be confused. Three vectors in $\mathbb{R}^2$ are always dependent, so Lemma 3 says nothing about triples of normals, Carter's lemma is unavailable at $k = 3$, and step (1) has no starting point, which makes steps (2) and (3) unreachable as well. Whether the conclusion of Theorem 11 holds in the plane is a separate question, and this paper decides it in neither direction.

Corollary 33 is not a counterexample to it: the shortest kernel elements recorded there have length 20, and a statement about elements of length 6 says nothing about elements of length 20. The planar tuples at which the statement has content are those carrying a kernel element of length 6 at all, and one such tuple, written out, is consistent with it. Take $a = (\sqrt{3}, 1)$, so that $\cos\theta_{0,1} = \frac{1}{2}$, $\cos\theta_{1,2} = \frac{\sqrt{3}}{2}$ and $\theta_{0,2} = \pi/2$: the triangle $O(a)$ has angles $\pi/2, \pi/3, \pi/6$ and $W(a)$ is the $(2, 3, 6)$ triangle group. Since $\theta_{0,1} = \pi/3$, the rotation $\rho_a(R_0 R_1)$ has order 3, so $(R_0 R_1)^3 \in \ker \rho_a$; this element has length 6 because $\langle R_0, R_1 \rangle$ is infinite dihedral in W_2, and with Lemma 10, which is proved for all $n \geq 2$, that gives $K(a) = 6$. The witnessing element lies in the rank-two standard parabolic $\langle R_0, R_1 \rangle$ and its pair of facets meets at angle $\pi/3$, which is what the conclusion of Theorem 11 asserts. One consistent tuple is evidence and not a proof, and no proof is offered.

4. The envelope in closed form

Theorem 14 (The envelope is rational)**.** *Let $c_{\Gamma_n}(x) = \sum_{C \text{ clique}} x^{|C|}$ be the clique polynomial of Γ_n. The growth series of W_n is rational and satisfies*

$$\frac{1}{W_n(t)} = c_{\Gamma_n}\Big(\frac{-t}{1+t}\Big).$$

Equivalently $W_n(t) = (1+t)^{n+1}/J_{n+1}(t)$, where

$$J_m = (1+t)\big(J_{m-1} - t\,J_{m-2}\big), \qquad J_0 = J_1 = 1.$$

One has $J_m = (1+t)^{\lfloor m/2 \rfloor} K_m$ with

$$K_0 = K_1 = 1, \qquad K_{2j} = K_{2j-1} - tK_{2j-2}, \qquad K_{2j+1} = (1+t)K_{2j} - tK_{2j-1}, \tag{2}$$

so that

$$W_n(t) = \frac{(1+t)^{\lfloor n/2 \rfloor + 1}}{K_{n+1}(t)}.$$

The roots of K_{n+1} in $(1/3, \infty)$ are exactly the numbers $\big(1 + 2\cos\frac{2k\pi}{n+3}\big)^{-1}$ for $1 \le k < (n+3)/3$; these are all of the positive roots of K_{n+1}, and the smallest of them is the radius of convergence of $W_n(t)$. Hence the sphere sizes $u_d = [t^d]\,W_n(t)$ satisfy $\limsup_{d\to\infty} u_d^{1/d} = r_n$, and the ball sizes $b_d = \sum_{e \le d} u_e$ satisfy $\lim_{d\to\infty} b_d^{1/d} = r_n$, with

$$r_n = 1 + 2\cos\frac{2\pi}{n+3}.$$

The interval is $(1/3, \infty)$ and not $(1/3, 1)$: roots above 1 occur, the first at $n = 4$ where K_5 has the root $1.8019\ldots$ alongside $1/r_4 = 0.4450\ldots$, and again at $n = 5$ (the root 1 exactly), $n = 7$ $(2.6180\ldots)$ and $n = 10$ $(3.4389\ldots)$. Only the smallest positive root enters r_n, so the conclusion is unaffected. The root set is certified exactly, as a polynomial identity over $\mathbb{Z}$, for $2 \le n \le 30$ (§10).

Proof. W_n is a right-angled Coxeter group, so its finite standard parabolics are exactly the cliques of Γ_n, each a group $(\mathbb{Z}/2)^{|C|}$ with growth polynomial $(1+t)^{|C|}$ [10]. Steinberg's formula [21, 3, 13] gives

$$\frac{1}{W_n(t^{-1})} = \sum_{C \text{ clique}} \frac{(-1)^{|C|}}{(1+t)^{|C|}} = c_{\Gamma_n}\Big(\frac{-1}{1+t}\Big).$$

Substituting $t \mapsto t^{-1}$ and simplifying $-1/(1+t^{-1}) = -t/(1+t)$ yields the displayed identity for $1/W_n(t)$. The substitution must be carried inside the clique polynomial; retaining the argument $-1/(1+t)$ gives a different rational function, already at $n = 2$.

The cliques of Γ_n are the independent sets of its complement, the path P_{n+1}, so $c_{\Gamma_n} = I_{n+1}$, the independence polynomial of the path on $n+1$ vertices, which obeys $I_m = I_{m-1} + xI_{m-2}$, $I_0 = 1$, $I_1 = 1+x$, and has the coefficient form $I_m(x) = \sum_k \binom{m+1-k}{k} x^k$. Put $J_m(t) = (1+t)^m I_m(-t/(1+t))$. Multiplying the recurrence by $(1+t)^m$ gives $J_m = (1+t)J_{m-1} - t(1+t)J_{m-2}$, which is the stated recurrence, with $J_0 = 1$ and $J_1 = (1+t)(1 - t/(1+t)) = 1$. By construction $1/W_n(t) = I_{n+1}(-t/(1+t)) = J_{n+1}(t)/(1+t)^{n+1}$. The factorisation $J_m = (1+t)^{\lfloor m/2 \rfloor} K_m$ and the two-step recurrences (2) follow by induction on m, splitting on the parity of m; cancelling $(1+t)^{\lfloor (n+1)/2 \rfloor}$ against $(1+t)^{n+1}$ leaves the exponent $\lceil (n+1)/2 \rceil = \lfloor n/2 \rfloor + 1$.

For the roots, fix $t > 0$ and solve the linear recurrence. Its characteristic equation is $\lambda^2 - (1+t)\lambda + t(1+t) = 0$, of discriminant $(1+t)(1-3t)$.

If $0 < t < 1/3$ both roots are real with $0 < \lambda_- < \lambda_+ < 1$, their sum being $1 + t \le 4/3$ and their product $t(1+t) < 4/9$. Writing $J_m = \alpha\lambda_+^m + \beta\lambda_-^m$, the initial conditions give $\alpha = (1-\lambda_-)/(\lambda_+ - \lambda_-) > 0$ and $\beta = (\lambda_+ - 1)/(\lambda_+ - \lambda_-) < 0$ with $\alpha + \beta = 1$. Then $J_m = \lambda_-^m(\alpha(\lambda_+/\lambda_-)^m + \beta)$, and the bracket is increasing in m with value 1 at $m = 0$, so $J_m > 0$. At $t = 1/3$ the root is double and $J_m = (1 + m/2)(2/3)^m > 0$. Hence J_m has no root in $(0, 1/3]$.

If $t > 1/3$ the roots are $\lambda_\pm = \varrho e^{\pm i\theta}$ with $\varrho = \sqrt{t(1+t)}$ and $\cos\theta = \sqrt{(1+t)/(4t)}$, so θ ranges over $(0, \pi/3)$ and is strictly increasing in t, because $(1+t)/(4t) = \frac{1}{4} + \frac{1}{4t}$ is strictly decreasing. Then $J_m = \varrho^m(\cos m\theta - c\sin m\theta)$, where $J_0 = 1$ is automatic and $J_1 = 1$ forces $c\varrho\sin\theta = (t-1)/2$, that is $c = (t-1)/\sqrt{(1+t)(3t-1)}$. Since $\sin^2\theta = (3t-1)/(4t)$, a direct computation gives $\cot 2\theta = (1-t)/\sqrt{(1+t)(3t-1)} = -c$. Therefore

$$J_m = 0 \iff \cot m\theta = c = -\cot 2\theta \iff \frac{\sin((m+2)\theta)}{\sin m\theta\,\sin 2\theta} = 0 \iff (m+2)\theta \in \pi\mathbb{Z}.$$

Both divisions in that chain are legitimate. For $\theta \in (0, \pi/3)$ one has $\sin 2\theta \neq 0$; and if $\sin m\theta = 0$ then $J_m = \varrho^m \cos m\theta = \pm\varrho^m \neq 0$, so such a θ is not a root and may be set aside before dividing.

With $m = n+1$ the roots are exactly $\theta = k\pi/(n+3)$ with $0 < \theta < \pi/3$, that is $1 \le k < (n+3)/3$. Inverting $\cos^2\theta = (1+t)/(4t)$ gives $t = 1/(4\cos^2\theta - 1) = 1/(1 + 2\cos 2\theta)$, so the corresponding t-values are $t_k = 1/(1 + 2\cos\frac{2k\pi}{n+3})$, and $1 + 2\cos\frac{2k\pi}{n+3} \in (0, 3)$ on that index range, so $t_k > 1/3$. Since $1 + t_k > 0$, each t_k is a root of K_{n+1} and not of the stripped factor $(1+t)^{\lfloor (n+1)/2 \rfloor}$. Conversely K_{n+1} has no root in $(0, 1/3]$ by the previous paragraph, so its roots in $(1/3, \infty)$, equivalently its positive roots, are exactly the t_k. As θ is strictly increasing in t, the smallest of them is t_1.

It remains to connect that root to the coefficients, which no step so far does: a root of the denominator is not by itself a singularity of the quotient, and a singularity of the quotient is not by itself the radius of convergence. Write R for the radius of convergence of $W_n(t) = \sum_{d \ge 0} u_d t^d$.

First, $0 < R < \infty$. The u_d are sphere sizes for the generating set $\{R_0, \ldots, R_n\}$, so $0 \le u_d \le (n+1)^d$ and $R \ge 1/(n+1)$. For $n \ge 2$ the vertices 0 and 1 of Γ_n are non-adjacent, so $\langle R_0, R_1 \rangle$ is infinite dihedral and W_n is infinite; hence no sphere is empty, since $u_d = 0$ would give $B_d = B_{d-1}$ and therefore $W_n = B_{d-1}$ finite. So $u_d \ge 1$ for all d, the series diverges at $t = 1$, and $R \le 1$.

Second, $R \ge t_1$. The coefficients u_d are non-negative, so by Pringsheim's theorem the point $t = R$ is a singularity of $W_n(t)$. A rational function is analytic off the zero set of any denominator representing it, so $K_{n+1}(R) = 0$. Since $R > 0$, the previous paragraph gives $R \in \{t_k\}$ and therefore $R \ge t_1$.

Third, $R \le t_1$. Here $K_{n+1}(t_1) = 0$ while the numerator satisfies $(1 + t_1)^{\lfloor n/2 \rfloor + 1} \neq 0$, because $t_1 > 1/3 > -1$. Cancelling the greatest common divisor of numerator and denominator cannot remove the factor $(t - t_1)$ from the denominator, since a common factor would have to vanish at t_1 and the numerator does not. So in lowest terms the denominator still vanishes at t_1, that is, W_n has a pole at t_1, and a power series cannot converge on a disc containing a pole of its sum. Hence $R \le t_1$.

Therefore $R = t_1 = 1/r_n$ with $r_n = 1 + 2\cos\frac{2\pi}{n+3}$, and by Cauchy–Hadamard $\limsup_d u_d^{1/d} = 1/R = r_n$. Since $r_n \ge r_2 = \varphi > 1$, the ball generating function $W_n(t)/(1-t)$ has the same radius R, and the ball sizes are submultiplicative, $b_{d+e} \le b_d b_e$, so Fekete's lemma upgrades the upper limit to a limit there: $\lim_d b_d^{1/d} = r_n$. □

Remark 15 (Not a path spectral radius). The number $2\cos\frac{2\pi}{n+3}$ is not the spectral radius of the path P_{n+1}, which is $2\cos\frac{\pi}{n+2}$; the two disagree for every $n \ge 2$ (at $n = 2$ they are $0.618\ldots$ and $1.414\ldots$). The angles produced by the proof are $k\pi/(n+3)$, and the growth rate uses $\cos 2\theta$ at $\theta = \pi/(n+3)$, not $\cos\theta$.

Remark 16 (The clique polynomial is a Fibonacci–Pell polynomial). The coefficient form $I_{n+1}(x) = \sum_k \binom{n+2-k}{k} x^k$ used above is the statement that the clique count of Γ_n in size k is $\binom{n+2-k}{k}$, and it identifies c_{Γ_n} with the Fibonacci–Pell polynomial P_{n+3}, where $P_0 = 0$, $P_1 = 1$ and $P_m = P_{m-1} + yP_{m-2}$: the two families satisfy the same three-term recursion, by Pascal's identity $\binom{n-k+2}{k} = \binom{n-k+1}{k} + \binom{n-k+1}{k-1}$, and agree at the bottom. This gives a second reading of the roots. The characteristic equation $x^2 = x + y$ has roots $\alpha, \beta = (1 \pm \sqrt{1+4y})/2$, the Binet formula yields $P_m(y) = (\alpha^m - \beta^m)/(\alpha - \beta)$, so $P_m(y) = 0$ iff $(\alpha/\beta)^m = 1$; setting $\alpha/\beta = e^{2\pi i j/m}$ gives $y_j = -1/(4\cos^2(\pi j/m))$. Translating back through the substitution of the theorem, $y = -t/(1+t)$, that is $t = -y/(1+y)$, and using $4\cos^2\vartheta - 1 = 1 + 2\cos 2\vartheta$, returns $t_j = 1/(4\cos^2(\pi j/m) - 1) = 1/(1 + 2\cos\frac{2\pi j}{n+3})$ at $m = n+3$, which are the roots located in the proof. The variant $y = -1/(1+t)$ would return $1 + 2\cos\frac{2\pi j}{n+3}$ in place of its reciprocal, that is the rate in place of the root; it is the substitution error the proof of Theorem 14 warns against, and it is harmless here only because the two lists are reciprocal. The Pascal step, the Binet formula and the translation identity are machine-checked (§10); the induction that assembles the Pascal step into the identification $c_{\Gamma_n} = P_{n+3}$ is a paper proof.

Corollary 17 (Asymptotic regime). *$r_n \to 3$ as $n \to \infty$, with*

$$3 - r_n = 4\sin^2\Big(\frac{\pi}{n+3}\Big) \sim \frac{4\pi^2}{(n+3)^2}.$$

Since $2\cos(2\pi/m)$ has degree $\phi(m)/2$ over $\mathbb{Q}$, where ϕ denotes Euler's totient, the rate r_n is rational or a quadratic surd exactly when $\phi(n+3) \le 4$, that is $n+3 \in \{5, 6, 8, 10, 12\}$; see Table 1, where φ denotes the golden ratio.

n	$W_n(t)$	r_n	algebraic form
2	$(1+t)^2/(1-t-t^2)$	$1.6180\ldots$	$\varphi = (1+\sqrt{5})/2$ (golden)
3	$(1+t)^2/(1-2t)$	2.0000	exactly 2
4	$(1+t)^3/(1-2t-t^2+t^3)$	$2.2470\ldots$	$1+2\cos(2\pi/7)$, cubic
5	$(1+t)^3/(1-3t+t^2+t^3)$	$2.4142\ldots$	$1+\sqrt{2}$ (silver)
6	$(1+t)^4/(1-3t+3t^3)$	$2.5321\ldots$	$1+2\cos(2\pi/9)$, cubic
7	$(1+t)^4/(1-4t+3t^2+2t^3-t^4)$	$2.6180\ldots$	$\varphi^2 = (3+\sqrt{5})/2$
8	$(1+t)^5/(1-4t+2t^2+5t^3-2t^4-t^5)$	$2.6825\ldots$	$1+2\cos(2\pi/11)$, degree 5
9	$(1+t)^5/(1-5t+6t^2+2t^3-4t^4)$	$2.7321\ldots$	$1+\sqrt{3}$
10	$(1+t)^6/(1-5t+5t^2+6t^3-7t^4-2t^5+t^6)$	$2.7709\ldots$	$1+2\cos(2\pi/13)$, degree 6

Table 1: The envelope $W_n(t) = (1+t)^{\lfloor n/2\rfloor+1}/K_{n+1}(t)$ and its growth rate $r_n = 1+2\cos(2\pi/(n+3))$. The recurrence (2), its agreement with the Steinberg expansion, the exactness of the division by $(1+t)^{\lfloor (n+1)/2\rfloor}$, and the identification of the smallest positive root of K_{n+1} with $1/r_n$ are certified for $2 \le n \le 30$ by the program of §10.

Remark 18 (The orthoscheme sequences in the OEIS). For pairwise distinct legs the layer counts $u_d(a)$ agree with the envelope as far as they have been computed, and the resulting sequences are recorded in the On-Line Encyclopedia of Integer Sequences: A397439 ($n = 3$), A397437 ($n = 4$), A396927 ($n = 5$) and A397438 ($n = 6$), with the $n = 7$ entry in preparation. Each of these is the coefficient sequence of the rational function $W_n(t)$ of Table 1, with dominant pole $1/r_n$. The planar entry A396406 is of a different nature: it records the deviated planar sequence, which agrees with $W_2(t)$ only up to degree 9 and whose generating function is transcendental over $\mathbb{Q}(x)$ [2]; see Corollary 33.

5. Rank-two relations

This section determines exactly when $\ker \rho_a$ meets a rank-two standard parabolic $\langle R_i, R_{i+1}\rangle$. Throughout it, the legs are positive rationals. By Lemma 1 the only pairs that can carry a nontrivial dihedral relation are the consecutive ones, and their cosines are

$$\cos\theta_{0,1} = \frac{a_2}{\sqrt{a_1^2+a_2^2}}, \qquad \cos\theta_{n-1,n} = \frac{a_{n-1}}{\sqrt{a_{n-1}^2+a_n^2}}, \qquad \cos\theta_{i,i+1} = \frac{a_i a_{i+2}}{\sqrt{(a_i^2+a_{i+1}^2)(a_{i+1}^2+a_{i+2}^2)}} \tag{3}$$

for $1 \le i \le n-2$. In each case $\cos^2\theta \in \mathbb{Q}$. By the extended Niven theorem, a consequence of Conway–Jones [7] (if $\theta \in \mathbb{Q}\pi$ and $\cos^2\theta \in \mathbb{Q}$ then $\cos\theta \in \{0, \pm\frac{1}{2}, \pm\frac{\sqrt{2}}{2}, \pm\frac{\sqrt{3}}{2}, \pm 1\}$), a relation $(R_i R_{i+1})^m = 1$ in $W(a)$ forces

$$\cos^2\theta_{i,i+1} \in \{0, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}, 1\}.$$

5.1 The two positional statistics

For $a \in \mathbb{Q}^n_{>0}$ set

$$\begin{aligned} e(a) &= \#\{i \in \{1, n-1\} : a_i = a_{i+1}\} \in \{0,1,2\}, \\ t(a) &= \#\{i : 1 \le i \le n-2,\ a_i = a_{i+1} = a_{i+2}\}. \end{aligned}$$

Thus e counts equal pairs at the two endpoint positions of the staircase and t counts interior triples of three consecutive equal legs. An equal pair sitting strictly inside the staircase, such as $a_2 = a_3$ in $(1,2,2,3)$ at $n = 4$, contributes to neither.

5.2 Boundary pairs

Lemma 19 (Boundary pairs)**.** *A boundary pair carries an admissible value of* $\cos^2\theta$ *if and only if the two legs involved are equal, in which case* $\cos^2\theta = \frac{1}{2}$ *and* $\theta = \pi/4$.

Proof. Take the pair $(0,1)$, so $\cos^2\theta = a_2^2/(a_1^2+a_2^2)$ by (3); the pair $(n-1,n)$ is symmetric. The value 0 forces $a_2 = 0$ and the value 1 forces $a_1 = 0$, both excluded by positivity. The value $\frac{1}{4}$ gives $4a_2^2 = a_1^2 + a_2^2$, that is $(a_1/a_2)^2 = 3$, and the value $\frac{3}{4}$ gives $(a_2/a_1)^2 = 3$; neither is possible over $\mathbb{Q}$. The value $\frac{1}{2}$ gives $a_1^2 = a_2^2$, that is $a_1 = a_2$, and conversely $a_1 = a_2$ gives $\cos\theta = 1/\sqrt{2}$. □

All five cases of Lemma 19 are formalised in Lean 4 (§10).

5.3 Interior pairs: three Diophantine quartics

At an interior pair write $x = a_i$, $y = a_{i+1}$, $z = a_{i+2}$, so that by (3)

$$\cos^2\theta_{i,i+1} = \frac{x^2 z^2}{(x^2+y^2)(y^2+z^2)}.$$

Requiring this to equal a constant c and setting $u = y^2$ gives the quadratic

$$c\,u^2 + c\,(x^2+z^2)\,u + (c-1)\,x^2 z^2 = 0, \tag{4}$$

which has a rational root only if its discriminant $\Delta = c^2(x^2+z^2)^2 - 4c(c-1)x^2z^2$ is the square of a rational. Substituting the three interior values of c and clearing the constant factors gives, in the two flanking legs $x = a_i$ and $z = a_{i+2}$, the Diophantine *quartics*

$$\mathcal{V}_{1/4}:\ x^4+14x^2z^2+z^4 = k^2, \qquad \mathcal{V}_{1/2}:\ x^4+6x^2z^2+z^4 = k^2, \qquad \mathcal{V}_{3/4}:\ 9x^4+30x^2z^2+9z^4 = k^2.$$

It matters that these are quartics in the legs and not conics in $X = x^2$, $Z = z^2$. Written in X and Z the three equations define conics with infinitely many rational points, among them $(X, Z) = (3, 2)$ on $X^2 + 6XZ + Z^2 = k^2$ with $k = 7$, and $(X, Z) = (2, 5)$ and $(5, 9)$ on the other two. Those points are irrelevant here precisely because X and Z must themselves be squares of legs.

Lemma 20. *Each of $\mathcal{V}_{1/4}$, $\mathcal{V}_{1/2}$, $\mathcal{V}_{3/4}$ has only trivial solutions over $\mathbb{Q}^2_{>0}$ in the legs (x, z): every rational point has $xz = 0$ or $x = z$.*

Proof. $\mathcal{V}_{1/2}$. The quartic $x^4 + 6x^2z^2 + z^4 = k^2$ is birationally equivalent to the elliptic curve $E_{1/2} : y^2 = x^3 - x$ (Mordell [19], §17). $E_{1/2}$ has Mordell–Weil rank 0 over $\mathbb{Q}$ by Fermat's classical descent (Cassels [6], Ch. II), and its full rational-point set is the 2-torsion $\{\infty, (0, 0), (\pm 1, 0)\}$, pulling back to $xz = 0$ or $x = z$.

$\mathcal{V}_{1/4}$. The Jacobian elliptic curve is $E_{1/4} : y^2 = x(x - 12)(x - 16)$, which is Cremona `24a1` (conductor 24, $j = 35\,152/9$, Mordell–Weil rank 0 over $\mathbb{Q}$, torsion $\mathbb{Z}/4 \oplus \mathbb{Z}/2$); see [9] and the LMFDB record [16]. Each of its eight rational points pulls back to $xz = 0$ or $x = z$.

$\mathcal{V}_{3/4}$. The quartic admits a standard conic-to-Weierstrass substitution and reduces to $E_{3/4}$: $y^2 = x(x - 300)(x - 1200)$, Cremona `72a2` (conductor 72, the same j-invariant $35\,152/9$ as $E_{1/4}$, of which it is a quadratic twist, also rank 0, torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$); see [9, 17]. Each torsion point again pulls back to a trivial quartic solution. □

Remark 21 (Provenance of the three rank-zero inputs). The three Mordell–Weil ranks were recomputed for this paper and are not taken on the authority of the tables. PARI/GP 2.17.4 returns matching lower and upper bounds 0 from a two-descent (`ellrank`) on each of $y^2 = x(x - 12)(x - 16)$, $y^2 = x^3 - x$ and $y^2 = x(x - 300)(x - 1200)$. Independently, `ellanalyticrank` returns analytic rank 0 on all three, with $L(E, 1) = 0.53912\ldots$, $0.65551\ldots$ and $0.97326\ldots$; nonvanishing of $L(E, 1)$ gives Mordell–Weil rank 0 again, by modularity together with the theorem of Kolyvagin. The same computation returns conductors 24, 32, 72, j-invariants $35\,152/9$, 1728, $35\,152/9$, and torsion subgroups $\mathbb{Z}/4 \oplus \mathbb{Z}/2$, $\mathbb{Z}/2 \oplus \mathbb{Z}/2$, $\mathbb{Z}/2 \oplus \mathbb{Z}/2$, and confirms that the third curve is the quadratic twist of the first by -3; all of these agree with the cited records. Two things are not re-derived here: the Cremona labels `24a1` and `72a2`, which are a naming convention and are used only as names, and the birational reductions of the three quartics to these curves, which are the standard ones cited in the proof.

Lemma 20 says nothing about the diagonal $x = z$, which is where the interior analysis has to be completed separately.

Lemma 22 (Interior pairs with equal flanking legs)**.** *Suppose $a_i = a_{i+2}$. Then the interior pair $(i, i + 1)$ carries an admissible value of $\cos^2\theta$ if and only if $a_i = a_{i+1} = a_{i+2}$, in which case $\cos^2\theta = \frac{1}{4}$ and $\theta = \pi/3$.*

Proof. With $x = z$ the cosine is $\cos\theta = x^2/(x^2 + y^2)$, a positive rational. The values 0 and 1 of $\cos^2\theta$ are excluded by positivity. The value $\frac{1}{2}$ requires $\cos\theta = 1/\sqrt{2}$ and the value $\frac{3}{4}$ requires $\cos\theta = \sqrt{3}/2$, neither rational. The value $\frac{1}{4}$ requires $\cos\theta = \frac{1}{2}$, that is $2x^2 = x^2 + y^2$, hence $y = x$; and conversely $x = y = z$ gives $\cos\theta = x^2/(2x^2) = \frac{1}{2}$. □

5.4 The classification

Theorem 23 (Rank-two classification)**.** *Let $n \geq 2$ and $a \in \mathbb{Q}^n_{>0}$. Then $\ker\rho_a$ meets some rank-two standard parabolic $\langle R_i, R_{i+1}\rangle$ nontrivially if and only if $e(a) + t(a) \geq 1$. More precisely:*

(i) if $t(a) \geq 1$, the interior pair at a triple has $\theta = \pi/3$ and $(R_iR_{i+1})^3 \in \ker\rho_a$ is an element of length 6;

(ii) if $t(a) = 0$ and $e(a) \geq 1$, the corresponding boundary pair has $\theta = \pi/4$ and $(R_i R_{i+1})^4 \in \ker \rho_a$ is an element of length 8, and no shorter element of any rank-two standard parabolic lies in $\ker \rho_a$;

(iii) if $e(a) = t(a) = 0$, no rank-two standard parabolic meets $\ker \rho_a$ nontrivially.

Proof. Since $\langle R_i, R_{i+1} \rangle$ is an infinite dihedral standard parabolic of W_n, a word $(R_i R_{i+1})^m$ is geodesic of length $2m$, and $\ker \rho_a$ meets that parabolic nontrivially exactly when $\theta_{i,i+1} \in \pi\mathbb{Q}$, in which case the shortest element is $(R_i R_{i+1})^m$ with $\theta_{i,i+1} = \pi r/m$ in lowest terms. Non-consecutive pairs generate finite parabolics of W_n itself and contribute nothing.

Suppose $\theta_{i,i+1} \in \pi\mathbb{Q}$ for some i. By the extended Niven theorem $\cos^2 \theta_{i,i+1} \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$. If the pair is a boundary pair, Lemma 19 gives $a_i = a_{i+1}$ with the pair at an endpoint, that is $e(a) \geq 1$, and $\theta = \pi/4$. If the pair is interior with flanking legs $x = a_i \neq a_{i+2} = z$, then (4) has the rational root $u = a_{i+1}^2$, so its discriminant is a rational square and (x, z) is a nontrivial point of one of $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$, contradicting Lemma 20. If the pair is interior with $a_i = a_{i+2}$, Lemma 22 gives $a_i = a_{i+1} = a_{i+2}$, that is $t(a) \geq 1$, and $\theta = \pi/3$. This proves (iii) and the "only if" direction.

Conversely, if $t(a) \geq 1$ then some interior pair has $\theta = \pi/3$ by Lemma 22, the dihedral image has order 6, and the shortest relator is $(R_i R_{i+1})^3$ of length 6: (i). If $t(a) = 0$ and $e(a) \geq 1$ then some boundary pair has $\theta = \pi/4$ by Lemma 19, the dihedral image has order 8, and the shortest relator in that parabolic is $(R_i R_{i+1})^4$ of length 8; by the previous paragraph no other consecutive pair contributes an element of $\ker \rho_a$, since a boundary pair would force $e \geq 1$ with $\theta = \pi/4$ again and an interior pair would force $t \geq 1$: (ii). □

Remark 24 (The rationality hypothesis is necessary). Theorem 23 is false over $\mathbb{R}$, and fails already at the endpoint pair. Take $n = 3$ and $a = (\sqrt{3}, 1, 5)$. By (3), $\cos^2 \theta_{0,1} = a_2^2/(a_1^2 + a_2^2) = 1/4$, so $\theta_{0,1} = \pi/3$ and $(R_0 R_1)^3 \in \ker \rho_a$ is an element of length 6, while $e(a) = t(a) = 0$. The step that breaks is the one marked in Lemma 19: over $\mathbb{Q}$ the value $\cos^2 \theta = 1/4$ forces $(a_1/a_2)^2 = 3$ and is excluded, and over $\mathbb{R}$ it is not. The same substitution $a_1 = \sqrt{3}\, a_2$ produces such a tuple in every dimension $n \geq 2$, so no reformulation of (e, t) repairs the statement over $\mathbb{R}$; what would be needed is a classification of the real tuples on which a consecutive cosine is a Niven value, which is a different problem and is not treated here. The hypothesis is used only through Lemma 19, Lemma 20 and Lemma 22.

Following the terminology of the three-dimensional analysis, call a leg tuple *Class C* when its coordinates are pairwise distinct. Class C tuples have $e = t = 0$, so Theorem 23(iii) applies, but so do tuples with an interior equal pair such as $(1, 2, 2, 3)$ and tuples with a non-adjacent repetition such as $(1, 2, 1, 3)$; the latter is precisely the case that Lemma 22, and not Lemma 20, has to handle.

Conjecture 25 (Class C faithfulness). *For every $n \geq 3$ and every $a \in \mathbb{Q}_{>0}^n$ with pairwise distinct coordinates, ρ_a is injective; equivalently $u_d(a) = [t^d]\, W_n(t)$ for every $d \geq 0$.*

Theorem 23(iii) is the rank-two half of Conjecture 25 and is proved. What is missing is that a nontrivial kernel element must lie in a rank-two parabolic. Theorem 11 supplies exactly that in length 6, which is the shortest length available by Lemma 10; in unbounded length the statement is false in general, the planar case $n = 2$ being a counterexample. There the two consecutive pairs are both boundary pairs, so for $a_1 \neq a_2$ Theorem 23(iii) applies and no rank-two parabolic meets $\ker \rho_a$; yet $\ker \rho_a \neq \{1\}$ by Proposition 32, and its shortest elements have length 20, arising as $w_1 w_2^{-1}$ from eight pairs of distinct geodesic words w_1, w_2 of length 10 using all three generators [2]. Conjecture 25 is verified computationally at $n = 3$ with legs $(1, 2, 3)$ through depth 10 and at $n = 4$ with legs $(1, 2, 3, 5)$ through depth 8 (§10).

5.5 A determinant identity, and what it does not give

The Lorentzian Gram matrix of the orthoscheme in staircase coordinates, with the standard $(n,1)$ endpoint correction, is the tridiagonal $(n+1)\times(n+1)$ matrix Q_n with

$$\begin{aligned} &Q_n[0,0]=1-a_1^2, && Q_n[i,i]=a_i^2+a_{i+1}^2\ (1\le i\le n-1), && Q_n[n,n]=1,\\ &Q_n[0,1]=a_2, && Q_n[i,i+1]=a_i\,a_{i+2}\ (1\le i\le n-2), && Q_n[n-1,n]=a_{n-1}. \end{aligned}$$

Lemma 26 (Uniform Schur-complement determinant identity)**.** *For every $n\ge 3$ and every positive tuple $(a_1,\dots,a_n)$,*

$$\det Q_n=-\prod_{i=1}^n a_i^2.$$

Proof. Let $D_k=\det Q_n[:k{+}1,:k{+}1]$ be the leading principal minor of order $k+1$. Tridiagonal expansion along the last row gives

$$D_k=Q_n[k,k]\,D_{k-1}-Q_n[k-1,k]^2\,D_{k-2}\qquad(1\le k\le n),\tag{†}$$

with $D_{-1}=1$. We claim

$$D_k=\Big(\prod_{i=1}^k a_i^2\Big)\Big(1-\sum_{i=1}^{k+1}a_i^2\Big)\qquad\text{for }1\le k\le n-1.\tag{$\star$}$$

Base. $D_0=1-a_1^2$ and $D_1=(1-a_1^2)(a_1^2+a_2^2)-a_2^2=a_1^2(1-a_1^2-a_2^2)$.

Interior step. For $2\le k\le n-1$ the recurrence (†) reads $D_k=(a_k^2+a_{k+1}^2)D_{k-1}-(a_{k-1}a_{k+1})^2D_{k-2}$. Introducing $S_{k-1}=\sum_{i=1}^{k-1}a_i^2$ and $P_{k-2}=\prod_{i=1}^{k-2}a_i^2$ and substituting ($\star$) reduces the step to the polynomial identity

$$\begin{aligned} P_{k-2}\,a_{k-1}^2\,a_k^2\,(1-S_{k-1}-a_k^2-a_{k+1}^2)=(a_k^2+a_{k+1}^2)\,&P_{k-2}\,a_{k-1}^2\,(1-S_{k-1}-a_k^2)\\ &-(a_{k-1}a_{k+1})^2\,P_{k-2}\,(1-S_{k-1}) \end{aligned}$$

in the five independent symbols $(a_{k-1},a_k,a_{k+1},S_{k-1},P_{k-2})$. It is independent of k and of n.

Final step. At $k=n$ the recurrence uses $Q_n[n,n]=1$ and $Q_n[n-1,n]^2=a_{n-1}^2$, giving $D_n=D_{n-1}-a_{n-1}^2D_{n-2}$. With $S_{n-1}=\sum_{i=1}^{n-1}a_i^2$ and $P_{n-2}=\prod_{i=1}^{n-2}a_i^2$,

$$D_{n-1}-a_{n-1}^2D_{n-2}=P_{n-2}\,a_{n-1}^2\big[(1-S_{n-1}-a_n^2)-(1-S_{n-1})\big]=-P_{n-2}\,a_{n-1}^2a_n^2=-\prod_{i=1}^n a_i^2.$$

The three identities close the induction for every $n\ge 3$, with no per-n verification. All of them, the induction that assembles them, and the boundary step are machine-checked in Lean 4 over an arbitrary commutative ring; no positivity, ordering or characteristic hypothesis is used (§10). □

Remark 27 (Why nonvanishing of $\det Q_n$ does not give faithfulness). An earlier version of this material argued from Lemma 26 that $\det Q_n\ne 0$, hence the form is nondegenerate, hence ρ_a is faithful by Tits' theorem. That argument is a non sequitur, in three independent ways, and the deduction is withdrawn.

First, it proves too much. The hypothesis of Lemma 26 is that the legs are positive and nothing more, so the conclusion $\det Q_n\ne 0$ holds at every positive tuple, including $(1,1,\dots,1)$. Were the inference valid, ρ_a would be faithful there too. It is not: breadth-first search at $n=3$ with legs $(1,1,1)$ gives layer counts $1,4,9,17,28,42$ against the envelope $1,4,9,18,36,72$, a deficit already at depth 3, in agreement with Theorem 29, which assigns $\mathrm{cd}_3 = 3$ to that tuple. The same computation at $n = 4$ with legs $(1,1,1,1)$ gives $1, 5, 14, 31, 59, 101$ against $1, 5, 14, 33, 75, 169$.

Second, Tits' theorem concerns the canonical geometric representation of an abstract Coxeter system, on the space spanned by the simple roots with the form $B(\alpha_i, \alpha_j) = -\cos(\pi/m_{ij})$ [3]. It is not a statement about ρ_a, which is a different representation, on $\mathbb{R}^n$ rather than $\mathbb{R}^{n+1}$ and affine rather than linear. Nor is Q_n a Coxeter Gram matrix: its diagonal is not identically 1, and $Q_n[0,0] = 1 - a_1^2$ is negative whenever $a_1 > 1$.

Third, run at $n = 2$ the argument returns a false answer. There $\det Q_2 \neq 0$ just as well, so it would give $u_d(a) = [t^d]W_2(t) = F_{d+3}$ for all d; in fact the two agree only up to $d = 9$ and differ at $d = 10$, where the orbit count is 225 against the envelope's 233, a deficit of 8 accounted for by eight coincidences between distinct words of length 10 [2].

6. The collision depth

The collision depth $\mathrm{cd}_n(a)$ of Definition 4 is defined at every positive leg tuple $a \in U$. By Corollary 9, $\mathrm{cd}_n(a) = K(a)/2$, and by Lemma 10, $K(a) \geq 6$. Conjecture 25 says exactly that $\mathrm{cd}_n = \infty$ for every Class C tuple. The value $K(a) = 6$ is decided by combining the two preceding sections; from here to the end of the section the legs are positive rationals, because the input from §5 is.

Corollary 28 (When $K(a) = 6$)**.** *Let $n \geq 3$ and $a \in \mathbb{Q}^n_{>0}$. Then $K(a) = 6$ if and only if $t(a) \geq 1$. In particular $K(a) \geq 8$, hence $\mathrm{cd}_n(a) \geq 4$, at every tuple with $t(a) = 0$.*

Proof. If $t(a) \geq 1$ then Theorem 23(i) exhibits an element of $\ker \rho_a$ of length 6, and $K(a) \geq 6$ by Lemma 10, so $K(a) = 6$.

Conversely let $K(a) = 6$. By Theorem 11 the witnessing element is $(R_i R_{i+1})^{\pm 3}$ with $\theta_{i,i+1} = \pi/3$, so $\cos^2 \theta_{i,i+1} = \frac{1}{4}$. If $(i, i+1)$ is a boundary pair, Lemma 19 says the only value in the Niven list attained there is $\frac{1}{2}$, so this is impossible. If it is an interior pair with $a_i \neq a_{i+2}$, then (4) holds with $c = \frac{1}{4}$ and with the rational root $u = a_{i+1}^2$, so its discriminant is a rational square and (a_i, a_{i+2}) is a point of $\mathcal{V}_{1/4}$ with $a_i a_{i+2} \neq 0$ and $a_i \neq a_{i+2}$, which Lemma 20 excludes. Only $\mathcal{V}_{1/4}$ can arise at this step: the other two quartics record $c = \frac{1}{2}$ and $c = \frac{3}{4}$, and $\theta_{i,i+1} = \pi/3$ gives $c = \frac{1}{4}$. If it is an interior pair with $a_i = a_{i+2}$, Lemma 22 gives $a_i = a_{i+1} = a_{i+2}$, that is $t(a) \geq 1$.

For the last statement, let $t(a) = 0$. Then $K(a) \neq 6$ by what precedes, $K(a) \geq 6$ by Lemma 10, and $K(a)$ is even or infinite by Corollary 9; hence $K(a) \geq 8$ and $\mathrm{cd}_n(a) \geq 4$. □

The next theorem describes cd_n for every leg multiplicity pattern. Although the multiplicity partition of $(a_1, \ldots, a_n)$ has $p(n)$ possible shapes, the two local positional statistics (e, t) already fix the collision depth wherever $e + t \geq 1$, at the two values 3 and 4; the pattern enters no further.

Theorem 29 (Collision depth, all $n \geq 3$)**.** *Let $n \geq 3$ and $a \in \mathbb{Q}^n_{>0}$, and write $e = e(a)$, $t = t(a)$. Then $\mathrm{cd}_n(a) \geq 3$, and the four mutually exclusive and exhaustive cases are*

$$\mathrm{cd}_n(a) \begin{cases} = 3 & \text{if } t \geq 1, \\ = 4 & \text{if } t = 0 \quad \text{and } e \geq 1, \\ = \infty & \text{if } e = t = 0 \quad \text{and } \rho_a \text{ is injective}, \\ \in \{4, 5, 6, \ldots\} & \text{if } e = t = 0 \quad \text{and } \rho_a \text{ is not injective}. \end{cases}$$

The first two cases are unconditional and determine $\mathrm{cd}_n(a)$ outright. The last two are a case split on a property that this paper does not decide for $n \geq 3$; what is asserted there is the stated value in the one case and the stated bound in the other.

Proof. By Corollary 9 and Lemma 10, $\mathrm{cd}_n(a) = K(a)/2$ and $K(a) \geq 6$; this gives $\mathrm{cd}_n(a) \geq 3$ in all four cases.

If $t \geq 1$ then $K(a) = 6$ by Corollary 28, so $\mathrm{cd}_n(a) = 3$.

If $t = 0$ and $e \geq 1$ then Theorem 23(ii) exhibits an element of $\ker \rho_a$ of length 8, so $K(a) \leq 8$, while Corollary 28 gives $K(a) \geq 8$. Hence $K(a) = 8$ and $\mathrm{cd}_n(a) = 4$.

Let $e = t = 0$. If ρ_a is injective then $\mathrm{cd}_n(a) = \infty$ by Corollary 8(i). If ρ_a is not injective then $K(a) < \infty$, so $\mathrm{cd}_n(a) = K(a)/2$ is finite, and $K(a) \geq 8$ by Corollary 28, so $4 \leq \mathrm{cd}_n(a) < \infty$. □

Remark 30 (What is and is not decided by (e,t)). Theorem 29 determines cd_n outright on the two strata $t \geq 1$ and $t = 0 < e$, with no residual ambiguity in the value. On the third stratum $e = t = 0$ it reduces the question to a single one, whether ρ_a is injective, and gives $\mathrm{cd}_n \geq 4$ if it is not. What Theorem 11 supplies is a lower bound on $K(a)$ off the triple stratum; it supplies no upper bound anywhere. The rank-two analysis of §5 controls the elements of $\ker \rho_a$ lying in a dihedral standard parabolic, Theorem 11 controls all elements of length 6, and beyond length 6 nothing in this paper controls the elements outside those parabolics.

Remark 31 (The run-length rule fails in $n \geq 4$). A natural but incorrect heuristic is to declare cd_n finite as soon as some consecutive equal pair appears, that is as soon as the run length $R = \max_i \#\{j : a_j = a_i\}$ is at least 2. At $n = 3$ this rule coincides with the (e,t) rule, because every consecutive equal pair in a three-term sequence is automatically at an endpoint. At $n \geq 4$ the two diverge: the orthoscheme $(1,2,2,3)$ has an interior equal pair, so $R = 2$ but $(e,t) = (0,0)$, and Theorem 29 places it on the stratum $e = t = 0$, its third and fourth cases, where the theorem asserts no finite collision depth; empirically its layer counts $1, 5, 14, 33, 75, 169, \ldots$ are identical to those of the Class C tuple $(1,2,3,5)$ through every computed depth, falsifying the run-length rule.

n	leg sequence	(e,t)	cd_n	first six layer counts
3	$(2,3,5)$	$(0,0)$	∞	$1,4,9,18,36,72$
3	$(1,1,2)$	$(1,0)$	4	$1,4,9,18,35,67$
3	$(1,1,1)$	$(2,1)$	3	$1,4,9,17,28,42$
4	$(1,2,3,5)$	$(0,0)$	∞	$1,5,14,33,75,169$
4	$(1,2,2,3)$	$(0,0)$	∞	$1,5,14,33,75,169$
4	$(1,1,2,3)$	$(1,0)$	4	$1,5,14,33,74,163$
4	$(1,1,2,2)$	$(2,0)$	4	$1,5,14,33,73,157$
4	$(1,1,1,2)$	$(1,1)$	3	$1,5,14,32,67,135$
5	$(1,2,3,5,7)$	$(0,0)$	∞	$1,6,20,54,136,334$
5	$(1,2,2,3,5)$	$(0,0)$	∞	$1,6,20,54,136,334$
5	$(1,1,2,3,5)$	$(1,0)$	4	$1,6,20,54,135,327$
5	$(2,1,1,1,2)$	$(0,1)$	3	$1,6,20,53,128,297$
6	$(1,1,2,3,2,2)$	$(2,0)$	4	$1,7,27,82,224,581$

Table 2: A small atlas of representative orthoschemes in $n = 3,4,5,6$ illustrating each attainable (e,t) class. The layer counts are exact-rational breadth-first search (§10); in every row the observed collision depth is the value predicted by Theorem 29, and the entries ∞ are observed agreement with the envelope as far as computed, not a proof. The two $n = 4$ rows with $(e,t) = (0,0)$ have identical counts through every computed depth despite differing in run structure.

The rows with $\mathrm{cd}_n = \infty$ saturate the upper bound $[t^d]\, W_n(t)$ as far as they are computed, as Conjecture 25 predicts; for $n = 3$ this reproduces $u_d = 9 \cdot 2^{d-2}$ for $d \geq 2$.

7. The plane, and why its argument does not extend

Proposition 32 (The plane)**.** *For $n = 2$ and every $a \in U$, ρ_a is not injective, so alternative (ii) of Corollary 8 holds.*

Proof. $W(a) \leq \mathrm{Isom}(\mathbb{R}^2) = \mathbb{R}^2 \rtimes O(2)$. The group $O(2)$ contains the abelian $SO(2)$ with index 2, so $\mathrm{Isom}(\mathbb{R}^2)$ is virtually solvable as an abstract group, hence amenable, and so is every subgroup of it; in particular $W(a)$ is amenable. The right-angled Coxeter group W_2 is not amenable: its defining graph Γ_2 is not complete multipartite, since its complement P_3 is connected and is not a disjoint union of cliques, and a right-angled Coxeter group whose defining graph is not complete multipartite contains a free subgroup of rank 2 [10]. An amenable group is not isomorphic to a non-amenable one, so ρ_a is not injective. □

Corollary 33 (The planar horizon)**.** *For $n = 2$ and every shape admissible in the sense of [2], the collision depth is $\mathrm{cd}_2 = 10$, equivalently $K(a) = 20$: $u_d(a) = F_{d+3} = [t^d]W_2(t)$ for $1 \leq d \leq 9$ (while $u_0 = 1$ and $F_3 = 2$), and at $d = 10$ the count is 225 against the envelope's 233. Past that horizon the sequence is* A396406*, with growth rate $\beta_2 = 1.4916\ldots < \varphi = r_2$, and its generating function is transcendental over $\mathbb{Q}(x)$ [2]. This is the only dimension in which the deviation has been computed past its onset.*

The isoceles right triangle $a_1 = a_2$ is excluded from that statement, and must be: there $e(a) = 1$, so Theorem 23(ii) gives an element of $\ker \rho_a$ of length 8 and hence $\mathrm{cd}_2 \leq 4$.

7.1 The scope of the ambient obstruction

Proposition 32 is an instance of one argument scheme: find a property of abstract groups that passes to subgroups, holds for the ambient group $\mathrm{Isom}(\mathbb{R}^n)$, and fails for W_n. Call such a property an *ambient obstruction* in dimension n. The scheme decides the alternative of Corollary 8 in the plane. The next theorem determines exactly when it can decide anything at all, and shows that in dimension $n \geq 3$ the specific property used in the plane is not merely unproved but false of the ambient group.

Theorem 34 (Ambient obstructions exist only in the plane)**.** *For $n \geq 2$ consider the statement*

$$(E_n): \qquad W_n \text{ is isomorphic to a subgroup of } \mathrm{Isom}(\mathbb{R}^n).$$

- *(i) If (E_n) fails, then ρ_a is not injective for every $a \in U$.*
- *(ii) If (E_n) holds, then every property of abstract groups that passes to subgroups and holds for $\mathrm{Isom}(\mathbb{R}^n)$ also holds for W_n. No ambient obstruction exists in dimension n, and no argument of that scheme can decide Problem 36 there.*
- *(iii) $\mathrm{Isom}(\mathbb{R}^n)$ is amenable as an abstract group if and only if $n \leq 2$, while W_n contains a free subgroup of rank 2 for every $n \geq 2$. Hence amenability is an ambient obstruction in dimension 2 and in no other; for $n \geq 3$ both groups contain free subgroups of rank 2.*

Proof. (i) If ρ_a were injective then $W_n \cong W(a) \leq \mathrm{Isom}(\mathbb{R}^n)$, which is (E_n).

(ii) Immediate from the definition: a subgroup-closed property of $\mathrm{Isom}(\mathbb{R}^n)$ holds for all its subgroups, and under (E_n) the group W_n is one of them.

(iii) Write $\mathrm{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$. For $n \leq 2$ the group $SO(n)$ is abelian of index 2 in $O(n)$, so $\mathbb{R}^n \rtimes SO(n)$ is solvable of index 2 in $\mathrm{Isom}(\mathbb{R}^n)$, which is therefore virtually solvable and amenable. For $n \geq 3$ the block embedding $SO(3) \leq SO(n) \leq \mathrm{Isom}(\mathbb{R}^n)$ and Hausdorff's free subgroup of rank 2 in $SO(3)$ [22, Thm. 2.1] make $\mathrm{Isom}(\mathbb{R}^n)$ non-amenable. For W_n: the complement of Γ_n is the path P_{n+1} on $n + 1 \geq 3$ vertices, which is connected and is not a clique, so Γ_n is not complete multipartite and W_n contains a free subgroup of rank 2 [10]. □

Remark 35 (Reading of Theorem 34)*.* Three consequences are worth separating.

First, the temptation to be resisted. One is tempted to argue that $\mathrm{Isom}(\mathbb{R}^n)$ is amenable because $O(n)$ is compact. Compactness gives amenability of $O(n)$ as a *topological* group, with respect to Haar measure, and that property does not pass to abstract subgroups. As an abstract group $O(n)$ is non-amenable for every $n \geq 3$, by the Hausdorff free subgroup that is the standard ingredient of the Banach–Tarski paradox. At $n = 2$ the argument survives only because $O(2)$ is virtually abelian.

Second, the amenability route is not repairable by replacing amenability with a different invariant. By Theorem 34(ii) the whole scheme is available in dimension n exactly when (E_n) fails, so any successful instance of it proves $\neg(E_n)$, whatever invariant it is dressed in.

Third, the scheme is strictly stronger than what Problem 36 asks. By Theorem 34(i), $\neg(E_n)$ gives non-injectivity of ρ_a at *every* leg tuple simultaneously, whereas Problem 36(a) asks only about the generic one. So an affirmative answer to Problem 37 below would close the scheme without saying anything about ρ_a, and a negative answer would settle Problem 36 outright. Either way the question is not a detail of the write-up; it is the exact content of the barrier.

8. What is proved, and the three open problems

8.1 What is proved

Table 3 lists the status of every numbered statement. The following are proved unconditionally: the dichotomy (Theorem 7); generic rigidity (Theorem 5), which makes "the generic n-orthoscheme growth" a well-defined object realised by a co-null set of shapes; rationality of the envelope with rate r_n (Theorem 14); the exclusion of length-six kernel elements outside the rank-two standard parabolics, at every positive leg tuple in every dimension $n \geq 3$ (Theorem 11); and the rank-two classification (Theorem 23). Together the last two determine the collision depth from the positional statistics (e, t) on the two strata $t \geq 1$ and $t = 0 < e$, and give $\mathrm{cd}_n \geq 3$ everywhere and $\mathrm{cd}_n \geq 4$ wherever $t = 0$ (Theorem 29).

What is not determined is cd_n on the third stratum $e = t = 0$, and there the question is exactly whether ρ_a is injective. That is settled here in the plane only, by Proposition 32, where the answer is no and the horizon is $\mathrm{cd}_2 = 10$. For $3 \leq n \leq 7$ the geometric growth has been computed to depth 13 to 16 and agrees with the envelope throughout that range. That agreement is consistent with both alternatives of Corollary 8: under (i) it would continue forever, and under (ii) it would end at a horizon beyond the computed range. Nothing here decides between them.

8.2 The three open problems

The three problems below are the whole of what this paper leaves open. Each is stated in full, with what is known about it. Only the first is stated as a conjecture, and the evidence on both sides of it is recorded below; for the other two we state a question and decline to guess a direction.

Problem 36 (The alternative in dimension $n \geq 3$)**.** *Fix $n \geq 3$.*

(a) Generic form. *Is $N_{\mathrm{gen}} = \bigcap_{a \in U} \ker \rho_a$ trivial? Equivalently, by Theorem 5, is ρ_a injective for every a in the co-null set U_{gen}; equivalently, by Corollary 8, is the growth series of the generic n-orthoscheme group the rational envelope $W_n(t)$ in every degree?*

(b) Arithmetic form. *Is ρ_a injective for every Class C tuple $a \in \mathbb{Q}^n_{>0}$? This is Conjecture 25.*

What is known. The two forms are related in one direction only. By Corollary 6, a single faithful tuple, rational or not, answers (a) affirmatively; so (b) implies (a). The converse is not automatic, because U_{gen} contains no tuple with algebraically dependent coordinates, hence no rational tuple: an affirmative answer to (a) says nothing directly about any given rational a. The rank-two half of (b) is proved, Theorem 23(iii). No soft upgrade of that half to the whole is available: faithfulness would follow if every nontrivial kernel element lay in a rank-two parabolic, and that implication is false, the plane being a counterexample, where for $a_1 \neq a_2$ no rank-two parabolic meets $\ker \rho_a$ while $\ker \rho_a$ is nontrivial with shortest elements of length 20 using all three generators [2]. The same counterexample is a warning about (b) itself: the $n = 2$ analogue of Conjecture 25 is false. The positive evidence is confined to $n \geq 3$ and is computational, the agreement to depths 13 to 16 recorded above, together with Theorem 11, which removes the shortest length at which a relation could have appeared. If the answer to (a) is affirmative for all $n \geq 3$, the plane is the unique dimension with non-rational orbit growth. If instead $\ker \rho_a$ is nontrivial in some dimension $n \geq 3$, the growth deviates below the envelope at $\mathrm{cd}_n(a) = K(a)/2 \geq 4$, and the arithmetic of the deviated series past that horizon is open even in the cases where the horizon can be located.

Problem 37 (The ambient embedding)**.** *Fix $n \geq 3$. Is W_n isomorphic to a subgroup of* $\mathrm{Isom}(\mathbb{R}^n)$*; that is, does statement* (E_n) *of Theorem 34 hold?*

What is known. (E_2) is false, by amenability, and that is the entire content of Proposition 32. For $n \geq 3$ the amenability route to $\neg(E_n)$ is closed: $\mathrm{Isom}(\mathbb{R}^n)$ contains a free subgroup of rank 2 and so does W_n (Theorem 34(iii)). The obvious finite-subgroup invariant does not separate them either: every finite subgroup of W_n is conjugate into one of the elementary abelian 2-groups on the cliques of Γ_n [10], hence has rank at most $\lceil (n+1)/2 \rceil \leq n$, while $O(n)$ contains $(\mathbb{Z}/2)^n$. By Theorem 34(ii) this single question governs the entire class of ambient obstructions: if (E_n) holds, no such obstruction exists in dimension n; if it fails, Problem 36 is answered negatively at every leg tuple at once. We do not conjecture a direction. Note that a positive answer would not touch Problem 36: an abstract embedding of W_n need not be by reflections in the facets of an orthoscheme, or by reflections at all.

Problem 38 (Finite presentation)**.** *Fix $n \geq 3$. Is the generic n-orthoscheme reflection group W_n/N_{gen} finitely presented?*

What is known. At $n = 2$ the answer is no: Isola and Piergallini [14] show that the generic triangle reflection group is not finitely presented, which gives non-injectivity in the plane independently of the amenability argument. Since W_n is finitely presented, an answer either way decides Problem 36(a): a negative answer refutes it, and hence also refutes Conjecture 25 by Corollary 6, while an affirmative answer is consistent with it and does not prove it. Problems 36 and 38 are therefore not independent, and the direction in which the plane answers Problem 38 is the direction that would refute Conjecture 25. We know of no argument in either direction for $n \geq 3$.

8.3 Further questions

Three further questions are open and are stated for the record.

(Q1) *Other diagrams.* Does an analogue of Theorem 29 hold for other families of simplicial reflection chambers in $\mathbb{R}^n$, for instance Coxeter orthoschemes with bent paths or simplices from non-path-shaped diagrams? The analysis here uses the right-angled-with-one-nontrivial-bond structure of the path diagram throughout.

(Q2) *Intermediate rates.* In every dimension $n \geq 3$, multiplicity patterns that are neither Class C nor saturated by triples produce growth rates strictly between two consecutive values of r_n. Whether these intermediate rates are algebraic over $\mathbb{Q}$ at all is open.

(Q3) *Beyond (e, t).* Theorem 29 shows that the collision depth is determined by the positional statistics (e, t). Is there a finer positional invariant that determines the entire sequence $u_d(a)$, not only its first deviation?

Statement	Status	Lean coverage, and what remains on paper
Lemma 1 (facet normals)	proved*	`OrthoschemeNormals`: the support bookkeeping, with the support pattern taken as a definition; the derivation of the normals from the geometry is on paper
Lemma 2 (facet intersections)	proved*	none
Lemma 3 (unique relation)	proved*	none
Theorem 5 (generic rigidity)	proved*	none
Corollary 6	proved*	none
Theorem 7 (dichotomy)	proved*	`EnvelopeDichotomy`: the horizon arithmetic and two abstract finset-counting lemmas; no word metric and no group
Corollaries 8, 9	proved*	none
Lemma 10 ($K \geq 6$)	proved*	none
Theorem 11 (length six is dihedral)	proved*, reviewed	`ReflectionTriple`: step (2) in full; steps (1) and (3) on paper. All $n \geq 3$; nonzero legs suffice, Rem. 12
Theorem 14 (envelope)	proved*	`EnvelopeDichotomy`, `NdimRate`: the recurrence ansatz, the root condition, the inversion $t = 1/(1 + 2\cos 2\theta)$. Steinberg, the clique count and the Pringsheim step are on paper
Corollary 17	proved*	`NdimRate`: the identity $3 - r_n = 4\sin^2(\pi/(n+3))$ in full; the asymptotic $\sim 4\pi^2/(n+3)^2$ on paper
Lemma 19 (boundary pairs)	proved*	`RankTwoExclusion`: the five Niven cases over $\mathbb{Q}$ in full; the cosine formula (3) and the extended Niven theorem on paper
Lemma 20 (three quartics)	proved*	none; rank-0 descents recomputed, not formalisable in current Mathlib
Lemma 22 (equal flanking legs)	proved*	none
Theorem 23 (rank-two classification)	proved*	none; interior half uses Lemma 20. Rational legs are necessary, Rem. 24
Lemma 26 ($\det Q_n$)	proved*	`OrthoschemeDet`: the recurrence-to-closed-form algebra, with the tridiagonal expansion as a hypothesis; the file contains no `Matrix.det`
Corollary 28 ($K = 6 \iff t \geq 1$)	proved*	none
Theorem 29, case $t \geq 1$	proved*	none; $\mathrm{cd}_n = 3$
Theorem 29, case $e \geq 1, t = 0$	proved*	none; $\mathrm{cd}_n = 4$, unconditional
Theorem 29, case $e = t = 0$	conditional	equivalent to Conj. 25
Conjecture 25 (Class C)	conjecture	checked to depth 10 ($n = 3$), 8 ($n = 4$)
Proposition 32 (the plane)	proved*	none
Theorem 34 (ambient obstruction)	proved*	none
Problem 36	open in both directions	
Problem 37 (ambient embedding)	open in both directions	false at $n = 2$
Problem 38 (finite presentation)	open in both directions	answered “no” at $n = 2$ [14]

Statement	Status	Lean coverage, and what remains on paper

Table 3: Status of the numbered statements. "Proved" means a complete written proof, and nothing more. The star marks formalisation debt: the statement is not verified in Lean 4, and in this paper *every* proved statement carries it, because no numbered statement is formalised end to end. The third column states exactly which steps the Lean files do cover; Table 4 gives the declarations and their hypotheses, and §10 lists what blocks the rest. "Reviewed" means that the written proof has been re-examined in full against a hostile reading, independently of its drafting; the hypothesis audit that review produced is Remark 12, and the status of the excluded dimension is Remark 13. The three rows at the bottom are the whole of what the paper leaves open.

9. Related work

Path Coxeter groups and growth-rate calculus. The closed form $r_n = 1 + 2\cos(2\pi/(n+3))$ is the signature of a linear, path-shaped Coxeter group whose Poincaré-series denominator reduces to a tridiagonal Chebyshev-type characteristic polynomial. Cannon and Wagreich [4], Floyd and Plotnick [12] and Parry [20] developed the growth-rate calculus of such groups, and Kellerhals and Perren [15] treat right-angled Coxeter growth systematically. Theorem 14 fits into that framework with parameter $m = n + 3$.

A Salem-type family on the surface-dynamics side. The family $\{1 + 2\cos(2\pi/m)\}_{m\geq 3}$ to which r_n belongs also arises in birational surface dynamics. Diller and Favre [11] introduced the dynamical-degree framework for bimeromorphic maps of surfaces; McMullen [18] showed that this family arises as dynamical degrees of automorphisms of blowups of $\mathbb{P}^2$ associated with the E_n T-shaped Coxeter graph, which is not the path graph occurring here; and Bedford and Kim [1] realised specific members through linear-fractional recurrences. The rate r_n lies in this family, but the right-corner orthoscheme reflection group is not known to be realised as a McMullen involution product on a rational surface, and the coincidence is at the level of the algebraic-number family only. A birational-geometry construction realising r_n as a dynamical degree on a $\mathbb{P}^2$-blowup would be of independent interest.

10. Certification

Orbit counts. The layer counts quoted above, the envelope coefficients and the Diophantine searches are produced by `code/zeta_probe/tools/ortho_cd` (Rust, exact arithmetic in $\mathbb{F}_p$ at two primes). It builds the $n+1$ facet reflections directly from the legs, enumerates the group by breadth-first search, and compares layer by layer against $(1+t)^{n+1}/J_{n+1}(t)$. In particular it produces the counts of Remark 27 at $(1, 1, 1)$ and $(1, 1, 1, 1)$ and at $n = 2$, confirms the (e, t) prediction on every tuple of Table 2 including the non-adjacent repetition $(1, 2, 1, 3)$, and searches the three quartics of Lemma 20 for nontrivial solutions with legs up to 300, finding none, while exhibiting rational points of the corresponding conics in (X, Z) to show that the restriction to squares is not cosmetic.

Short kernel elements. Theorem 11 is a proof, not a computation, but it was first tested by exhaustive search and the search is worth recording, since it also tests Lemma 10 and the (e, t) rule of Corollary 28. The program `code/zeta_probe/tools/ortho_len6` (Rust, exact integer arithmetic in homogeneous coordinates, no floating point) enumerates the reduced words of length at most 3 in W_n, computes their images under ρ_a, and inspects every collision; a nontrivial element of $\ker \rho_a$ of length at most 6 exists exactly when a collision does, and its length and support are read off by the deletion condition for graph products. Every positive rational tuple scales to a primitive integer tuple and $\rho_a(w) = 1$ is scale invariant, so integer boxes cover the rational case. The boxes run are

$$n = 3,\ \text{legs} \le 200; \quad n = 4,\ \le 40; \quad n = 5,\ \le 14; \quad n = 6,\ \le 8; \quad n = 7,\ \le 5; \quad n = 8,\ \le 4,$$

comprising 9 954 322 primitive tuples in total. In every one of them: no element of $\ker \rho_a$ has length below 6; no element of length 6 has support of size 3 or more; and an element of length 6 occurs if and only if $t(a) \ge 1$, in which case it is $(R_i R_{i+1})^{\pm 3}$ at a triple of equal consecutive legs. The per-dimension counts are in the tool's `README.md`.

The three elliptic curves. The rank, torsion, conductor and j-invariant of each of the three Jacobians of Lemma 20 were recomputed with PARI/GP 2.17.4, by `ellrank` and, independently, `ellanalyticrank`; the outputs and what they do and do not establish are in Remark 21.

The envelope. Theorem 14 is certified by `code/zeta_probe/tools/racg_envelope` (Rust, exact integer polynomial arithmetic), which for $2 \le n \le 30$ checks that the recurrence for J_m and the Steinberg expansion agree coefficient by coefficient, performs the division by $(1+t)^{\lfloor (n+1)/2 \rfloor}$ exactly, and matches the smallest positive root of K_{n+1} against $1/\big(1+2\cos\frac{2\pi}{n+3}\big)$. Independently, for $2 \le n \le 6$ it enumerates W_n by breadth-first search in the canonical geometric representation, whose matrices are integral here because the bilinear form takes only the values $1, 0, -1$ and which is faithful by Tits' theorem [3]; the sphere sizes agree with the series coefficients to depths 9 to 14. The same program records two checks on the shape of the answer: the variant substitution $-1/(1+t)$ in place of $-t/(1+t)$ gives a different rational function already at $n = 2$, and the polynomial $1 - 2t - 3t^2 + 4t^3 - t^4$ is not the $n = 7$ denominator, being the negated reciprocal of $1 - 4t + 3t^2 + 2t^3 - t^4$ and predicting 6 elements at depth 1 in a group with 8 generators.

The root statement of Theorem 14 is certified separately and exactly, with no floating point in the decisive step, by `code/zeta_probe/tools/verify_scripts/ortho_envelope_roots.py`. For $2 \le n \le 30$ it checks, over $\mathbb{Z}$: the identity $J_m = (1+t)^{\lfloor m/2 \rfloor} K_m$; the agreement of the Steinberg expansion with the J recurrence; the exactness of the division; the printed denominators of Table 1; and the root identification, in the form that the reciprocal polynomial of K_{n+1} equals the monic integer polynomial whose roots are the numbers $1+2\cos(2k\pi/m)$ for $1 \le k \le \lfloor (m-1)/2 \rfloor$, $k \ne m/3$, that polynomial being built from $2T_m(x/2) - 2$ rather than from any numerical evaluation of a cosine. The count of real roots in $(1/3, \infty)$ is then obtained by Sturm sequences over $\mathbb{Q}$ and matched against $\#\{k : 1 \le k < m/3\}$, and the same method places the smallest root at $k = 1$. The four roots at or above 1 quoted after the theorem are confirmed to be roots of K_{n+1} by exact rational bracketing. The script also verifies the substitution of Remark 16 symbolically, in both its correct and its erroneous form. As a cross-check only, the roots are recomputed numerically at 30 and at 60 decimal digits and agree with the closed forms to better than 10^{-20} at both precisions. The dominant-pole step itself is not a computation and is not certified by any script; it is the paper argument in the proof of Theorem 14.

Lean 4. Six files in `lean/with_mathlib/` elaborate against Mathlib with no `sorry` and no `native_decide`, and the 52 declarations named below exist under exactly those names. That is the whole of what the Lean files certify, and it is less than it may appear: *no numbered statement of this paper is formalised end to end*, so no statement of this paper carries the label VERIFIED. Table 4 says, per file, what the declarations establish and what they take as hypotheses. The gap in every row is the same one, and it is not incidental: Mathlib has Coxeter systems and their length function (`Mathlib.GroupTheory.Coxeter.Length`) but no growth series, no Steinberg identity, no standard parabolics, and no graph products, so the objects these statements are about cannot yet be written down.

File (declarations)	Declarations; what they establish; what they assume
`OrthoschemeNormals` (7) Lemma 1	`supp_disjoint`, `mul_eq_zero_of_disjoint`, `sum_eq_zero_of_disjoint`, `dot_zero_one`, `dot_interior`, `dot_last`, `orthogonality_pattern`. *Establishes:* for the finite sets $\mathrm{supp}(n, j)$ equal to $\{1\}$, $\{j, j+1\}$, $\{n\}$, indices differing by at least 2 give disjoint sets; vectors supported on disjoint sets have vanishing coordinatewise product and vanishing sum; the three scalar expressions l_2, $-l_j l_{j+2}$, $-l_{n-1}$ are nonzero for positive legs. *Assumes:* the support pattern is a *definition* in the file, not a consequence of the orthoscheme geometry. The step "the facet opposite V_j has normal $l_{j+1}e_j - l_j e_{j+1}$", which is the content of Lemma 1, is on paper. No Euclidean space and no facet occur in the file.
`EnvelopeDichotomy` (12) Theorem 7, Theorem 14, Remark 15	`two_mul_lt_of_lt_horizon`, `split_lengths`, `card_image_lt_of_collision`, `card_lt_of_escapes`, `cos_three_term`, `sin_three_term`, `J_recurrence`, `cot_add_cot_eq_zero_iff`, `J_eq_zero_iff`, `t_of_theta`, `rate_of_theta`, `rate_ne_path_spectral_radius`. *Establishes:* the horizon arithmetic $d < \lceil K/2 \rceil \Rightarrow 2d < K$ and the parity split of K; two finite-set counting lemmas, one for a collision inside a finset and one for a point of a finset whose image escapes a covered target, each giving a strict drop in cardinality; that the ansatz $\varrho^m(\cos m\theta - c \sin m\theta)$ satisfies the recurrence of (2); the equivalence $\cot A + \cot B = 0 \iff \sin(A + B) = 0$ and its consequence for the ansatz; the inversion $t(1 + 2\cos 2\theta) = 1$; and that $2\cos(2\pi/5) \neq 2\cos(\pi/4)$. *Assumes:* there is no word metric, no group and no clique polynomial in the file. The two counting lemmas are about an arbitrary finset and an arbitrary map; that the spheres of G and G/N are such finsets and such a map is the paper proof of Theorem 7. The Steinberg identity, the clique count, and the Pringsheim step of Theorem 14 do not appear.
`NdimRate` (7) Remark 16, Corollary 17	`sq_eq`, `pell_binet`, `root_translation`, `rate_residual`, `three_sub_r`, `pascal_step`, `coeff_pascal`. *Establishes:* the Binet formula in division-free form $(\alpha - \beta)P_m = \alpha^m - \beta^m$ for any sequence satisfying the Pell recurrence; the identity $4\cos^2\vartheta - 1 = 1 + 2\cos 2\vartheta$; and, in full and with the paper's definition of r_n, the identity $3 - r_n = 4\sin^2(\pi/(n+3))$ of Corollary 17; Pascal's rule in the index shape the paper uses. *Assumes:* the induction assembling `coeff_pascal` into $c_{\Gamma_n} = P_{n+3}$ is on paper, as is the identification of P_m's roots with the y_j; the asymptotic $3 - r_n \sim 4\pi^2/(n+3)^2$ is not in the file.
`RankTwoExclusion` (10) Lemma 19	`not_isSquare_three_int`, `sq_ne_three`, `niven_zero`, `niven_one`, `niven_quarter`, `niven_three_quarter`, `niven_half`, `no_length_two`, `no_length_three_boundary`, `boundary_no_relation`. *Establishes:* 3 is not a square in $\mathbb{Z}$ nor in $\mathbb{Q}$; and, for positive rationals a, b, each of the five equations $b^2 = c(a^2 + b^2)$ with $c \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ is impossible except $c = \frac{1}{2}$, which forces $a = b$. This is the arithmetic of Lemma 19 in full. *Assumes:* that $\cos^2\theta_{0,1} = a_2^2/(a_1^2 + a_2^2)$, that is (3), is on paper; so is the extended Niven theorem, which restricts c to those five values. Mathlib's `NumberTheory.Niven` covers $\cos\theta \in \mathbb{Q}$ only, not the $\cos^2\theta \in \mathbb{Q}$ form used here. The interior cases are absent.
`OrthoschemeDet` (3) Lemma 26	`minor_closed_form`, `det_Q`, `det_Q_of_recurrence`. *Establishes:* over any commutative ring, a sequence D with the two stated initial values and the stated three-term recurrence satisfies $D_k = \left(\prod_{i \le k} a_i^2\right)\left(1 - \sum_{i \le k+1} a_i^2\right)$, and a quantity D_{top} with the stated boundary relation equals $-\prod_i a_i^2$. *Assumes:* the file contains no `Matrix` and no `Matrix.det`. That the leading principal minors of Q_n obey that recurrence, and that $\det Q_n$ is given by that boundary relation, are hypotheses of the Lean statements and are proved on paper. So the file certifies the algebra of the tridiagonal expansion, not the determinant identity.

File (declarations)	Declarations; what they establish; what they assume
`ReflectionTriple` (13) Theorem 11, step (2)	`inner_self_ne_zero'`, `eq_of_sub_eq_sub`, `eq_zero_of_sub_eq_self`, `refl_self`, `refl_of_inner_eq_zero`, `refl_add`, `refl_smul`, `refl_neg`, `refl_sub`, `refl_involutive`, `refl_injective`, `inner_refl_left`, `pairwise_orth_of_comp_eq_neg`. *Establishes:* in an arbitrary real inner product space, with the reflection defined from the inner product and with no dimension hypothesis, the reflection calculus, and then the rigidity statement: if u, v, w are nonzero, v, w are independent, and $s_u s_v s_w = -\mathrm{id}$, then u, v, w are pairwise orthogonal. This is step (2) of the proof of Theorem 11 in full, and it is the only full step of a paper proof among the six files. *Assumes:* nothing beyond the hypotheses stated. The reduction of Theorem 11 to that configuration, namely the word combinatorics in W_n, Carter's rank lemma and the affine argument of step (3), is on paper.

Table 4: The six Lean 4 files, by scope. Each row names every declaration cited from that file, what those declarations prove, and what they take as given. No row covers a numbered statement of this paper end to end; consequently no statement of the paper is labelled VERIFIED in Table 3, and the star there marks the resulting formalisation debt.

`NdimRate.lean`, `OrthoschemeDet.lean` and `ReflectionTriple.lean` end with `#print axioms`, one line per theorem of those three files. Of the twenty-three declarations audited, nineteen report exactly the standard axioms `propext`, `Classical.choice`, `Quot.sound`, and four report strictly fewer, with `pascal_step` depending on no axioms at all. The remaining three files carry the same `#print axioms` lines but their output has not been tabulated here.

What is not formalised, and what blocks it. The following are paper proofs with no Lean counterpart at all: Lemma 2, Lemma 3, Lemma 10, Lemma 20, Lemma 22, Theorem 5, Theorem 23, Corollary 28, Theorem 29, Proposition 32 and Theorem 34, together with Corollaries 6, 8, 9 and 33. Four obstructions account for all of them.

(a) *No growth series.* Mathlib has Coxeter systems and their length function, but no sphere or ball counting function, no Poincaré series, no Steinberg identity, no standard parabolics and no graph products. Everything about $W_n(t)$, u_d, cd_n and $K(N)$ waits on that. This blocks Theorem 7, Theorem 14, Corollaries 8, 9, 28, Lemma 10 and Theorem 29.

(b) *No Pringsheim theorem.* The dominant-pole step added to the proof of Theorem 14 needs the statement that a power series with non-negative coefficients has a singularity at its radius of convergence. Mathlib has no such result.

(c) *No Mordell–Weil ranks.* Mathlib has Weierstrass curves but no descent and no rank computation, so Lemma 20, and with it the interior half of Theorem 23, cannot be stated as anything but an axiom. The three ranks were recomputed with PARI/GP instead (Remark 21).

(d) *No amenability, and no extended Niven.* Mathlib has no amenable-group API, which blocks Proposition 32 and Theorem 34; and its Niven theorem is the $\cos\theta \in \mathbb{Q}$ form, not the $\cos^2\theta \in \mathbb{Q}$ form of Conway–Jones used in §5.

Of these, (b) is the only one that a self-contained formalisation could plausibly discharge without new Mathlib theory, and it is the one whose absence the previous version of this section concealed by calling the step "proved on paper" while the paper did not contain it. That step is now written out.

A. Decomposition into atoms

Table 3 records a status per statement; this appendix records the dependency structure behind it. Every numbered statement of the paper appears exactly once. The *label* column carries one of: PROVED (a complete written proof and nothing more), CONDITIONAL (proved modulo a stated open input), CONJECTURE, DEFINITION, OPEN, RECORD (a computation or a bibliographic provenance note, not a mathematical claim of this paper). A star on PROVED marks formalisation debt, that is, a statement not verified in Lean 4; here the star is on every proved row, since no statement of the paper is formalised end to end (Table 4). No row carries VERIFIED. The *depends on* column lists the atoms used in the proof, not every atom mentioned; "lit." marks an external input. The dependency relation is acyclic, and Theorem 29 (A29) is the target atom.

ID	Statement	Label	Depends on	Status
A1	Lem. 1: the facet normals, and orthogonality exactly off the diagonal band	proved*	—	done
A2	Lem. 2: the facet hyperplanes are distinct and meet as stated	proved*	A1	done
A3	Lem. 3: the only linear relation among the normals	proved*	A1	done
A4	Def. 4: the collision depth $\mathrm{cd}_n(a)$	definition	—	done
A5	Thm. 5: generic rigidity of $\ker \rho_a$ on a co-null set	proved*	A1, A3	done
A6	Cor. 6: one faithful shape answers the generic question	proved*	A5	done
A7	Thm. 7: spheres agree below $\lceil K(N)/2 \rceil$ and drop there	proved*	—	done
A8	Cor. 8: the envelope alternative	proved*	A7	done
A9	Cor. 9: $\mathrm{cd}_n(a) = K(\ker \rho_a)/2$	proved*	A4, A7	done
A10	Lem. 10: no kernel element of length 4, so $K \geq 6$	proved*	A1, A2	done
A11	Thm. 11: every length-six kernel element is dihedral, $n \geq 3$	proved*	A1, A2, A3, lit. (Carter)	done, reviewed
A12	Rem. 12: where each hypothesis of A11 is used	proved*	A11	done
A13	Rem. 13: A11 is undecided at $n = 2$, in either direction	open	A11	open
A14	Thm. 14: $W_n(t)$ rational, its positive roots, and r_n	proved*	A1, lit. (Steinberg)	done
A15	Rem. 15: $r_n - 1$ is not the spectral radius of P_{n+1}	proved*	A14	done
A16	Rem. 16: $c_{\Gamma_n} = P_{n+3}$, and the roots a second way	proved*	A14	done
A17	Cor. 17: $3 - r_n = 4\sin^2(\pi/(n+3))$	proved*	A14	done
A18	Rem. 18: the four OEIS entries match $W_n(t)$ as far as computed	heuristic	A14	done
A19	Lem. 19: a boundary pair closes iff its two legs are equal	proved*	A1, lit. (Conway–Jones)	done
A20	Lem. 20: the three quartics have only trivial points	proved*	lit. (rank 0: `24a1`, `72a2`, $y^2 = x^3 - x$)	done
A21	Rem. 21: the three ranks recomputed in PARI/GP	record	A20	done
A22	Lem. 22: an interior pair with $a_i = a_{i+2}$ closes iff the triple is equal	proved*	A1, lit. (Conway–Jones)	done
A23	Thm. 23: rank-two relations exist iff $e + t \geq 1$, for $a \in \mathbb{Q}^n_{>0}$	proved*	A19, A20, A22	done

ID	Statement	Label	Depends on	Status
A24	Rem. 24: A23 fails over $\mathbb{R}$ at $(\sqrt{3}, 1, 5)$	proved*	A19, A23	done
A25	Conj. 25: ρ_a is injective on Class C tuples, $n \geq 3$	conjecture	A23	open
A26	Lem. 26: $\det Q_n = -\prod_i a_i^2$	proved*	A1	done, side
A27	Rem. 27: A26 does not imply faithfulness	proved*	A26	done, side
A28	Cor. 28: $K(a) = 6$ iff $t(a) \geq 1$	proved*	A10, A11, A19, A20, A22	done
A29	Thm. 29: cd_n from (e, t); target atom	proved* / conditional	A9, A10, A23, A28	done on $t \geq 1$ and $t = 0 < e$; open on $e = t = 0$
A30	Rem. 30: what A29 does and does not decide	proved*	A29	done
A31	Rem. 31: the run-length rule fails at $n \geq 4$	heuristic	A29	done
A32	Prop. 32: ρ_a is not injective at $n = 2$	proved*	A8, lit. (amenability)	done
A33	Cor. 33: the planar horizon is $\mathrm{cd}_2 = 10$	proved*	A32, lit. (companion paper)	done
A34	Thm. 34: ambient obstructions exist only at $n = 2$	proved*	lit. (free subgroups of $SO(3)$)	done
A35	Rem. 35: reading of A34	proved*	A34	done
A36	Prob. 36: is ρ_a injective for $n \geq 3$?	open	A6, A25, A29	open
A37	Prob. 37: does W_n embed in $\mathrm{Isom}(\mathbb{R}^n)$?	open	A34	open
A38	Prob. 38: is the generic n-orthoscheme group finitely presented?	open	A36, lit. [14]	open

Table 5: The 38 numbered statements as atoms, with labels, dependencies and status. Twenty-two are theorem-like: twenty-one carry PROVED in full and A29 carries it on two of its four cases. All twenty-two carry the formalisation star, so the debt count is 22 and the count of statements at VERIFIED is 0. Two atoms, A26 and A27, are marked *side*: they correct an argument used in an earlier version of this material and are not on any path to A29. Every other atom lies on a path to A29 or is one of the three open problems that A29 leaves.

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